

Inference on Optimal Policy Values and Other Irregular Functionals via Smoothing

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Abstract

Constructing confidence intervals for the value of an optimal treatment policy is an important problem in causal inference. Insight into the optimal policy value can guide the development of reward-maximizing, individualized treatment regimes. However, because the functional that defines the optimal value is non-differentiable, standard semi-parametric approaches for performing inference fail to be directly applicable. Existing approaches for handling this non-differentiability fall roughly into two camps. In one camp are estimators based on constructing smooth approximations of the optimal value. These approaches are computationally lightweight, but typically place unrealistic parametric assumptions on outcome regressions. In another camp are approaches that directly de-bias the non-smooth objective. These approaches don't place parametric assumptions on nuisance functions, but they either require the computation of intractably-many nuisance estimates, assume unrealistic L^∞ nuisance convergence rates, or make strong margin assumptions that prohibit non-response to a treatment. In this paper, we revisit the problem of constructing smooth approximations of non-differentiable functionals. By carefully controlling first-order bias and second-order remainders, we show that a softmax smoothing-based estimator can be used to estimate parameters that are specified as a maximum of scores involving nuisance components. In particular, this includes the value of the optimal treatment policy as a special case. Our estimator obtains $\sqrt{n}$ convergence rates, avoids parametric restrictions/unrealistic margin assumptions, and is often statistically efficient.

1 Introduction

Individualized treatment policies, or mappings from a person's observed characteristics to an assigned treatment, are deployed in everyday decision-making pipelines across disparate domains. For instance, individualized policies are deployed in medicine to maximize the expected outcome or well-being of a patient [Qian and Murphy, 2011, Xu et al., 2022]. Likewise, such policies play an important role in advertising in order to maximize patron engagement and interaction [Bottou et al., 2013]. Perhaps due to the practical importance of developing individualized policies, a vast literature has developed that focuses on using observational data to generate a treatment policy with a high *value*, i.e. a large average outcome recipient marginalized over covariates [Athey and Wager, 2021].

However, because developing an individualized treatment policy may be costly or time consuming, a learner may first want to perform inference on the *value of the optimal treatment policy*. Letting $Y(1), Y(0)$ for now represent an individual's potential outcomes under treatment and control and $\pi(X) \in \{0, 1\}$ denote an assigned binary treatment for user with covariates X , the value of the optimal treatment policy is given by

$$V^* := \max_{\pi} \mathbb{E}[Y(\pi(X))].$$

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By definition, an optimal treatment policy attains the maximum possible expected outcome or reward of any possible treatment regime. By constructing confidence intervals around this value, the learner can make informed decisions about the viability of investing resources in constructing a highly-individualized treatment policy. As a concrete example, if a doctor notices that the upper confidence bound on the optimal policy value is not much higher than the value of a baseline treatment strategy, he may not be inclined to update his diagnostic policy. On the other hand, if the lower confidence bound is much larger than the baseline policy’s value, he may want to invest in the development of improved individualized diagnostic criteria, say using modern ML methods.

At face value, it may seem that building confidence intervals for the optimal policy value should be strictly easier than learning a good policy. However, this is generally not the case. The optimal policy value functional, which is defined as a maximum over an infinite class of treatment policies, is non-differentiable. Even in the binary treatment setting, the optimal policy value, which can be re-written in terms of the outcome regression $Q^*(a, x) = \mathbb{E}[Y \mid X = x, A = a]$ as

$$V^* = \mathbb{E}[\max\{Q^*(0, X), Q^*(1, X)\}]$$

still involves a non-differentiable maximum. Standard semi-parametric theory, which forms the backbone for doubly-robust and de-biased machine learning approaches to functional estimation, requires the Gateaux differentiability of the target functional, and thus breaks down when there are non-responders to treatment — that is when $\mathbb{P}(Q^*(1, X) = Q^*(0, X)) > 0$. To address this, researchers have needed to develop analyses that are tailor-made for this specific problem instance.

The existing literature on estimating the optimal policy value has largely split into two camps. One camp involves replacing the non-differentiable maximum $a \mapsto \max\{0, a\}$ with a smooth surrogate — typically either a *soft-max function* $a \mapsto \frac{ae^{\beta a}}{1+e^{\beta a}}$ or a *soft-plus function* $a \mapsto \beta^{-1} \log(1+e^{\beta a})$ [Goldberg et al., 2014, Chen et al., 2023, Park, 2024]. The degree of smoothing (given through β) can then be tuned to guarantee asymptotic convergence to optimal value. However, these existing smoothing approaches all assume a parametric form for the conditional average treatment effect (CATE) $Q^*(1, x) - Q^*(0, x)$, which may not be realistic in practice. In the second camp, authors have opted instead to study de-biasing approaches for the base, non-differentiable objective [Semenova, 2023, Luedtke and Van Der Laan, 2016]. These approaches avoid placing semi-parametric restrictions on the CATE, but either require estimating a number of nuisances proportional to the sample size or require the nuisance estimates to converge to ground truth in the supremum norm, which is unrealistic when modern ML methods are employed. There is thus a fundamental tradeoff in existing approaches between computational tractability and robustness to misspecification. The question considered in this paper is as follows: is it possible to have the “best of both worlds” and obtain a computationally efficient estimator that avoids unpalatable, parametric assumptions on the blip effects?

1.1 Our Contributions

We show that, by carefully combining soft-max smoothing with first-order de-biasing, one can construct confidence intervals for not only the value of the optimal treatment policy, but also for other generic irregular functionals as well. Our approach is computationally lightweight, robust to nuisance estimation errors, and does not require parametric restrictions on outcome regressions. Our generic approach is to first explicitly define Neyman orthogonal scores for a soft-max smoothed objective and then show how to select smoothing parameters to obtain asymptotic normality around the true parameter V^* . A section-by-section enumeration of contributions is as follows.

1. In Section 3, we show how to use softmax smoothing to estimate the value of the optimal treatment policy. In more detail, we approximate $V^* := \mathbb{E}[\max_{\ell}^N \{Q^*(\ell, X)\}]$ by a smoothed solution V^{β} to a Neyman orthogonal moment:

$$V^{\beta} := \mathbb{E} \left[\underbrace{\mathbf{sm}_{\ell}^{\beta} \{Q^*(\ell, X)\}}_{\text{softmax approximation}} + \underbrace{\sum_{\ell} \alpha_{\ell}^{\beta}(A, X)(Y - Q^*(A, X))}_{\text{first-order de-biasing}} \right]$$

where $\alpha_{\ell}^{\beta}(a, x)$ are nuisance functions to be defined in the sequel. We then show how to select the smoothing parameter β in terms of the sample size and the smoothness of the sub-optimality gaps

$\Delta_k := \max_{\ell} \{Q^*(\ell, X)\} - Q^*(k, X)$ near zero in order to obtain asymptotic normality around V^* . In particular, when the reward maximizing action is unique with probability one, the influence function of our estimator is precisely the efficient influence function noted in Luedtke and Van Der Laan [2016].

2. In Section 4, we consider instead the estimation of the generalized irregular parameter

$$V^* := \mathbb{E} \left[\max_{\ell} \{\psi_{\ell}(X, g_{\ell}^*)\} \right],$$

where $\psi_{\ell}(x; g_{\ell})$ are generic scores and the true nuisance g_{ℓ}^* is a regression. As in Section 3, we also show (a) how to construct Neyman orthogonal moments and (b) how to choose smoothing parameters in order to obtain asymptotic normality. As an example, we revisit the work of Levis et al. [2023] and apply our technique in constructing confidence intervals for Balke and Pearl bounds on the average treatment effect. Our result generalizes that of Levis et al. [2023] as it allows the maximizing index to be non-unique with arbitrarily large probability.

As noted above, throughout our work, we assume that the maximizing set $\arg \max_{\ell} \{\psi_{\ell}(X, g_{\ell}^*)\}$ can have more than one element with positive probability. In the case of performing inference on the optimal policy value, this is equivalent to saying that we allow $\mathbb{P}(\tau^*(X) = 0) > 0$ in general, where $\tau^*(x) := Q^*(1, x) - Q^*(0, x)$ denotes the CATE. The density assumption on the sub-optimal indices is also mild, and in this same simplified setting would reduce to $|\tau^*(X)|$ admitting density of the form $O(x^{\delta-1})$ in a region $(0, c)$ for some $\delta, c > 0$. Overall, we show that soft-max smoothing is a viable approach for estimating irregular functionals involving nuisance components even when no parametric substructure is assumed.

1.2 Related Work

Inference on Optimal Policy Value: Most immediately related to our work are papers that focus on estimating the value of the optimal treatment policy. Several works consider smoothing approaches to construct a “surrogate” objective for the optimal policy value. For instance, Goldberg et al. [2014] consider a dynamic (two-step) treatment regime and show how to use a “soft-plus” function $(a, b) \mapsto \frac{1}{\beta} \log(e^{\beta a} + e^{\beta b})$ to approximate $\max\{a, b\}$. However, the authors assume that the underlying Q-functions/blip effects are linear, thus limiting the applicability of their result. Further, the soft-plus approximation has non-negligible bias at $a = b = 0$, and thus lacks applicability when the probability of a treatment recipient being a non-responder is zero. More closely related to our work, Chen et al. [2023] also consider a smoothed objective in the dynamic treatment regime. By using soft-max smoothing instead of soft-plus smoothing, the authors are able to design estimating equations that allow for asymptotically normal estimates of structural parameters under common margin assumptions that allow a positive probability of non-responders. However, the authors still assume there is a known, parametric form for the blip effects, which may be undesirable in practice. Further, the score constructed by the authors is *not* Neyman orthogonal with respect to the blip effects, and thus is not applicable when we remove semi-parametric restrictions. By carefully constructing an appropriate Neyman orthogonal smoothed score and balancing tradeoffs between bias and second order errors, we are able to remove these parametric restrictions, which allows for realistic, non-parametric convergence rates for nuisance estimates. We also mention the work of Park [2024], who also studies the use of soft-max smoothing in estimating the value of the optimal treatment policy in two treatment settings. Curiously, the authors actually get slower than $\sqrt{n}$ convergence rates for their estimates, obtaining confidence intervals of width $\sqrt{\beta_n^2/n}$ where β_n is a smoothing parameter tending towards infinity. Our results, on the other hand, not only obtain $\sqrt{n}$ convergence rates, but also apply to other irregular functionals of interest. In Appendix E, we show that for deterministic, increasing choices of smoothing parameters, the variance of the author’s estimator tends towards zero as the sample size tends towards infinity.

Other authors have opted to study de-biasing approaches for the base, non-differentiable objective. For instance, Semenova [2023] analyze a first-order de-biased estimator for the optimal policy value. However, this estimator requires that nuisances be estimated at $o(n^{-1/4})$ rates in the L^{∞} norm, which is a very strict requirement when off-the-shelf ML estimators like boosted trees are used and no parametric restrictions are placed on nuisance functions. Further, they require that there is a unique maximizing index, which prohibits having non-responders with a positive probability. Laber et al. [2014] also take

a non-smoothing approach to estimating the optimal policy value in a dynamic regime. However, not only do the authors again require parametric structure on the blip effects, but they additionally construct pessimistic confidence intervals in regions of non-regularity (i.e. regions of non-responders) via a union bound and bootstrapping methods. Likewise, Chakraborty et al. [2013] propose a method based on m -out-of- n bootstrapping to build confidence intervals on structural parameters. These results are not applicable in our setting, as we make no semi-parametric restrictions on the blip effects. Most relevant to this work are the contributions of Luedtke and Van Der Laan [2016], who consider the estimation of the optimal policy value for both one-stage and dynamic treatment regimes. Instead of performing smoothing, the authors propose an estimator that averages a de-biasing term over many copies of nuisance estimates. This work, importantly, does not place any parametric restrictions on the blip effects/Q-functions, and the authors show that their estimator is semi-parametric efficient when there is no probability of non-responders. However, the estimator involves producing a number of nuisance estimates that continually grows with the sample size, thus making the applicability of the estimator untenable. Our smoothing-based estimator is not only efficient for regular laws, but also requires computing only a small, constant number of nuisances. Thus, we view it as a computationally-viable alternative to the estimator of Luedtke and Van Der Laan [2016].

Policy Learning: In the causal inference and reinforcement learning (RL) literature, a large body of work has developed related to learning an optimal or near-optimal individualized treatment policy. These include approaches based on Q-learning [Zhang et al., 2012, Watkins and Dayan, 1992, Watkins et al., 1989, Qian and Murphy, 2011, Zhao et al., 2009, 2011, Moodie et al., 2012], A-learning (or advantage learning) [Shi et al., 2024, Murphy, 2003, Moodie et al., 2007, Robins, 2004], empirical risk minimization [Athey and Wager, 2021, Luedtke and Chambaz, 2020, Kallus and Zhou, 2018], and more. We particularly emphasize works relating to empirical risk minimization [Athey and Wager, 2021, Kitagawa and Tetenov, 2018, Kitagawa et al., 2022], which describe how to learn policies within a generally infinite class as to obtain low *regret*, which is defined as the difference in value between the best policy in the class and the learned policy [Manski, 2004, Hirano and Porter, 2009]. We also note works on learning optimal policies in partially-identified settings [Olea et al., 2023, Kitagawa et al., 2023, Pu and Zhang, 2021], which may be of particular importance when one does not believe commonly-made conditional exogeneity assumptions. We note that the algorithms in the works listed above do not allow the learner to perform statistical inference on the value of the optimal treatment policy. Also relevant to our work are algorithms from the RL literature based on soft-Q learning [Garg et al., 2021, Nachum et al., 2017, Schulman et al., 2017], which use entropy regularization to learn a near-optimal Boltzmann policy. Again, these algorithms do not provide a means for a learner to perform inference on the value of the optimal policy.

Smoothing and Estimation of Irregular Functionals: Smoothing, the primary technique considered in this paper, has appeared in other sub-fields of statistics to aid in the estimation of irregular parameters. For instance, Levis et al. [2023] leverage soft-plus smoothing to construct confidence intervals for Balke and Pearl bounds on partially-identified average treatment effects. For any fixed $\beta > 0$, the authors show that their soft-plus estimates approach a smoothed version of the Balke and Pearl bounds. However, they do not show how to tune β as a function of the sample size and smoothness of conditional average treatment effect in order to obtain convergence to the true Balke and Pearl bounds. Instead, the authors instead use the bound $\frac{1}{\beta} \log(\sum v_i) \leq \max\{v_1, \dots, v_k\} + \frac{\log k}{\beta}$ to appropriately widen confidence intervals, which is pessimistic. Zhang et al. [2024] consider the problem of building confidence sets containing the argument minimizer of a vector given noisy observations. To accomplish this, the authors use softmax sampling via the exponential mechanism from the differential privacy literature. We again mention the aforementioned work by Semenova [2023], who provide an estimator for computing the maximum of a finite collection of regressions involving a nuisance component. The author’s results are applicable to many irregular parameters such as upper and lower bounds on quantiles and Balke and Pearl bounds. However, these estimators still require that nuisances are learned $o(n^{-1/4})$ rates in the L^∞ norm.

Causal Inference: Lastly, we touch on tools from the causal inference and semi-parametric statistics that are relevant to our work [Kosorok, 2008, Pearl, 2009, Van der Vaart, 2000, Tsiatis, 2006, Hernán and Robins, 2010]. In semi-parametric statistics, a learner aims to perform inference on a low-dimensional

structural parameters (average causal derivatives, average treatment effects) in the presence of unknown, generally infinite-dimensional nuisance functions (propensity scores, outcome regressions). Naive plug-in estimators are sensitive to misestimation in nuisance components, and thus we leverage de-biased/double machine learning (DML) methods [Chernozhukov et al., 2018, Neyman, 1979, Van Der Laan and Rubin, 2006, Van der Laan et al., 2007, 2011], which perform first-order corrections to estimators involving a nuisance component, to construct our confidence intervals. These methods are commonly leveraged in both causal inference [Wager and Athey, 2018, Athey et al., 2019, Runge et al., 2023] and causal machine learning [Foster and Syrgkanis, 2023, Whitehouse et al., 2024, Lan et al., 2025]. We also mention work on automatic de-biasing methods [Athey et al., 2018, Chernozhukov et al., 2022c,a,b], which allow for the direct estimation of nuisance parameters with unknown or complicated forms through minimizing a “Riesz Loss.”

2 Problem Formulation and Background

In this section, we discuss preliminaries needed throughout the rest of the paper. In paper, we discuss smoothing approaches for constructing confidence intervals for the value of the optimal treatment policy and other irregular functionals. While the case of performing inference on the optimal policy value is a special case of the more general problem, we describe it separately due to its importance in the literature.

Before jumping into exposition, we describe notation that will be used throughout. We assume the learner will have access to i.i.d. generalized observations of the form $Z \in \mathcal{Z}$, and let P_Z denote the distribution of Z . We assume the nested structure of observables $X \subset W \subset Z$ where X generally represents covariates and W an extended set of features. We assume these variables take values in generic measurable spaces $\mathcal{X}$ and $\mathcal{W}$ respectively, and we let P_X and P_W denote their respective distributions. As an example, one may have $Z = (X, A, Y)$ where $X \in \mathbb{R}^d$ are covariates, $A \in \{0, 1\}$ is a binary treatment, and $Y \in \mathbb{R}$ is an outcome. We let $\mathbb{E}[\cdot]$ and $\mathbb{P}(\cdot)$ denote expectation and probability respectively over all sources of randomness. For arbitrary independent random variables U and V with distributions P_U and P_V and an arbitrary function $m(u, v)$, we let $\mathbb{E}_V[m(U, V)] := \int m(U, v)P_V(dv) = \mathbb{E}_{U,V}[m(U, V) | U]$. In this case $\mathbb{E}[m(U, V)] = \mathbb{E}_{U,V}[m(U, V)] = \int \int m(u, v)P_U(du)P_V(dv)$. Given samples $Z_1, \dots, Z_n \sim P_Z$, we let $\mathbb{P}_n := \frac{1}{n} \sum_{m=1}^n \delta_{Z_m}$ denote the corresponding empirical distribution, $\mathbb{P}_n f(Z) := \frac{1}{n} \sum_m f(Z_m)$, and $\mathbb{G}_n := \sqrt{n}(\mathbb{P}_n - \mathbb{E}_Z)$.

For a given integer $N > 0$, we let $[N] := \{1, 2, \dots, N\}$. For $p \in [1, \infty)$, a random variable $Z \in \mathcal{Z}$ with distribution P_Z , and a function $f : \mathcal{Z} \rightarrow \mathbb{R}^d$, we let $\|f\|_{L^p(P_Z)} := (\mathbb{E}_Z \|f(Z)\|_p^p)^{1/p}$, where $\|a\|_p := (\sum_i |a_i|^p)^{1/p}$ for $a \in \mathbb{R}^d$. For $p = \infty$, we let $\|f\|_{L^\infty(P_Z)} := \text{ess sup } |f(Z)|$. We define the L^p space of functions by $L^p(P_Z) := \{f : \mathcal{Z} \rightarrow \mathbb{R}^d \text{ s.t. } \|f\|_{L^p(P_Z)} < \infty\}$. The dimension d of the range of functions will either be clear from context or made explicit.

Given a Banach space B , a mapping $T : B \rightarrow \mathbb{R}^d$, and elements $f^*, g \in H$, we define the first Gateaux derivative of T at f^* in the direction g as $D_f T(f^*)(g) := \frac{\partial}{\partial t} T(f^* + tg)|_{t=0}$. We likewise define the second Gateaux derivative as $D_f^2 T(f^*)(g) := \frac{\partial^2}{\partial t^2} T(f^* + tg)|_{t=0}$. If we have a bi-variate map $T : B \times B \rightarrow \mathbb{R}^d$ and $f_1^*, f_2^*, g_1, g_2 \in B$, we let the cross Gateaux derivative be defined as $D_{f_1, f_2} T(f_1^*, f_2^*)(g_1, g_2) := \frac{\partial^2}{\partial s \partial t} T(f_1^* + tg_1, f_2^* + sg_2)|_{t=s=0}$. If we have function $f : \mathbb{R}^d \rightarrow \mathbb{R}$, we let $\partial_i f(u)$ denote its i th partial derivative, $\nabla_u f(u)$ its gradient, and $\nabla_u^2 f(u)$ its Hessian. We will generally write $\partial_i f(g(x)) := \partial_i f(u)|_{u=g(x)}$, $\nabla_u f(g(x)) := \nabla_u f(u)|_{u=g(x)}$, and $\nabla_u^2 f(g(x)) = \nabla_u^2 f(u)|_{u=g(x)}$ for convenience unless otherwise noted. Lastly, for functions $f, g : \mathcal{Z} \rightarrow \mathbb{R}^d$, we define the interval between f and g as the bracket $[f, g] := \{h : \mathcal{Z} \rightarrow \mathbb{R} : \exists \lambda : \mathcal{X} \rightarrow [0, 1] \text{ s.t. } h(z) = \lambda(z)f(z) + (1 - \lambda(z))g(z)\}$.

2.1 Formulation for the Value of the Optimal Treatment Policy

In the setting, we assume observations are of the form $Z = (X, A, Y)$, where $X \in \mathcal{X}$ represents an individual’s features or covariates, $A \in [N]$ represents a treatment, and $Y \in \mathbb{R}$ represents an observed outcome. We make the following assumptions.

Assumption 1. *We assume the learner has access to i.i.d. samples $Z = (X, A, Y)$ satisfying the following conditions:*

1. (Consistency) There are potential outcomes $(Y(a) : a \in [N])$ such that $Y = Y(A)$.
2. (Conditional Ignorability) We assume that the potential outcomes $(Y(a) : a \in [N])$ are conditionally independent of treatment A given covariates X , i.e. $Y(a) \perp\!\!\!\perp A \mid X$ for each $a \in [N]$.
3. (Strong Positivity) Let $p^*(a \mid x) := \mathbb{P}(A = a \mid X = x)$ represent the propensity score. The propensity scores are bounded strictly away from 0 and 1 — i.e. there is $\eta > 0$ such that, for any action $\ell \in [N]$ and $x \in \mathcal{X}$,

$$\eta \leq p^*(\ell \mid x) \leq 1 - \eta.$$

In this setting, by a treatment policy, we mean a measurable mapping $\pi : \mathcal{X} \rightarrow [N]$ taking an individual's covariates to a treatment. Straightforward manipulation yields that the value of the optimal treatment policy can be equivalently characterized as

$$V^* = \mathbb{E} \left[\max_{\ell} \{Q^*(\ell, X)\} \right]$$

where again $Q^*(a, x) := \mathbb{E}[Y \mid X = x, A = a]$ is the outcome regression/Q-function. In the case of binary treatments $A = 0$ and $A = 1$, we will let $\tau^*(x) := Q^*(1, x) - Q^*(0, x)$ be the conditional average treatment effect (CATE). We also define the “naively de-biased” score

$$\Psi^*(Z; Q, \alpha) := \max_{\ell} \{Q(\ell, x)\} + \sum_{\ell=1}^N \alpha_{\ell}(a, x)(y - Q(a, x)), \quad (1)$$

where the true nuisance is $\alpha_k^*(a, x) := \mathfrak{s}_k(Q^*(\cdot, X)) \frac{\mathbb{1}\{a=k\}}{p^*(k|x)}$, where for $u \in \mathbb{R}^N$

$$\mathfrak{s}_k(u) \equiv \mathfrak{s}_k(u_1, \dots, u_N) := \frac{\mathbb{1}\{k \in \arg \max_{\ell} u\}}{|\arg \max_{\ell}(u)|}. \quad (2)$$

We note that when the distribution of Z is *regular*, meaning that

$$\mathbb{P} \left(\left| \arg \max_{\ell} Q_{\ell}^*(\ell, X) \right| > 1 \right) = 0,$$

we have the identity $\mathfrak{s}_k(Q^*(\cdot, X)) = \mathbb{1}\{k = \arg \max_{\ell} Q^*(\ell, X)\}$. In this case, $\Psi^*(Z; Q^*, \alpha^*)$ is precisely the efficient influence function, as argued in Luedtke and Van Der Laan [2016]. Efficiency theory has not been established for irregular laws, and in fact Hirano and Porter [2012] show that regular estimators do not exist for general irregular functionals.

2.2 General Irregular Functionals

More broadly, we derive Neyman orthogonal scores and asymptotic linearity results for a variety of irregular statistical parameters. We assume that observations take the generic form Z and that we have $X \subset W \subset Z$ for a generalized set of covariates X and an extended set of features W . Here, we adopt the general framework considered in Semenova [2023], which focuses on estimating the expected point-wise maximum over a class of functions involving nuisance components. This framework will not only apply to performing inference on the value of the optimal treatment policy but also many other examples. In more detail, the goal is to perform inference on the maximum of the scores:

$$V^* := \mathbb{E} \left[\max_{\ell} \psi_{\ell}(X; g_{\ell}^*) \right] = \mathbb{E} [\max \psi(X; g^*)],$$

where for each $k \in [N]$, ψ_k is a score depending on the data vector Z and nuisance component $g_k : \mathcal{W} \rightarrow \mathbb{R}^{d_k}$, and $g_k^*(w) \in L^2(P_W)$ is some true, unknown nuisance function. We let $\psi(X; g) := (\psi_1(X; g_1), \dots, \psi_N(X; g_N))$ for convenience throughout, where $g = (g_1, \dots, g_N)$. In the example of the value of the optimal treatment policy, for each $k \in [N]$, we have $W = (A, X)$, $g_k^*(w) = Q^*(a, x) = \mathbb{E}[Y \mid X = x, A = a]$, $\psi_k(x; g_k) = g_k(k, x)$,

and $\zeta_k^*(a, x) = \frac{\mathbb{1}\{a=k\}}{p^*(k|x)}$. We describe how to instantiate this framework with respect to another example, conditional Balke and Pearl bounds due to Levis et al. [2023], in Section 4.

We emphasize that, unlike the results of Semenova [2023], our contributions in Section 4 will allow for an arbitrarily large probability of the argument maximizer having more than one element. Furthermore, our results will only rely on L^2 convergence rates for nuisance estimates, not strong L^∞ rates. In the sequel, we make the following assumption on the scores $\psi_1(x; g_1), \dots, \psi_N(x; g_N)$.

Assumption 2. *For any $k \in [N]$, we assume that the score $\psi_k(x; g_k)$ satisfies several properties, listed below.*

1. *We assume there is a Z -measurable random variable $U_k \in \mathbb{R}^{d_k}$ such that $g_k^*(w) = \mathbb{E}[U_k \mid W = w]$.*
2. *The first Gateaux derivative of ψ_k is sufficiently regular at g_k^* to admit a conditional Riesz representer, i.e. there is a square-integrable function $\zeta_k^* : \mathcal{W} \rightarrow \mathbb{R}^{d_k}$ such that*

$$D_{g_k} \psi_k(X; g_k^*)(\omega) = \mathbb{E}[\zeta_k^*(W)^\top \omega(W) \mid X]$$

for all $\omega \in L^2(P_W)$.

3. *For any $\nu \in L^2(P_W)$ and $x \in \mathcal{X}$, the function $\ell_{\nu, x} : \mathbb{R} \rightarrow \mathbb{R}^N$ given by*

$$\ell_{\nu, x}(\epsilon) := \psi(x; g^* + \epsilon \nu)$$

is twice continuously differentiable in ϵ . Further, we assume for any $\nu \in L^2(P_W)$ and $C > 0$, there is a random variable $F_{\nu, C} \geq 0$ satisfying $\mathbb{E}F_{\nu, C} < \infty$ and

$$\sup_{|\epsilon| \leq C} \|\partial_\epsilon \ell_{\nu, x}(\epsilon)\|_2, \sup_{|\epsilon| \leq C} \|\partial_\epsilon^2 \ell_{\nu, x}(\epsilon)\|_2 \leq F_{\nu, C} \quad \text{almost surely.}$$

In words, first part of the assumption just states that each g_k^* is a regression of some random variables U_k onto the extended set of covariates W . This is trivially satisfied for the value of the optimal treatment policy, in which case $g_k^*(a, x) = \mathbb{E}[Y \mid X = x, A = a]$ for $k \in [N]$. The second assumption just states that the scores are sufficiently regular to admit a Riesz representer conditional on X . We have $\zeta_k^*(a, x) = \frac{\mathbb{1}\{a=k\}}{p^*(k|x)}$ for the optimal policy value. Finally, the third assumption says that each score is twice continuously differentiable with appropriately dominated derivatives for any fixed direction ν . This latter condition, while technical, is to allow to interchange of differentiation and expectation at several points in the sequel. Given $\psi_a(x; Q) = Q(a, x)$ is affine for the optimal policy value, we see that the third assumption is trivially satisfied. Important again will be the limiting score Ψ^* defined in Equation (1), except with α_k^* instead defined as $\alpha_k^*(w) = \mathfrak{s}_k(\psi(X; g^*))\zeta_k^*(W)$.

2.3 Smoothing and Density Conditions

We conclude by discussing the smoothing functions and density conditions considered in the sequel. Throughout this section, we will always use the value of the optimal treatment policy in the setting of a binary treatment as a motivating example. Nonetheless, our smoothing results are more general, applying to settings with more than two treatments and other irregular parameters of interest as well.

In the setting of a binary treatment, we can characterize the value of the optimal treatment policy as

$$V^* = \mathbb{E}[\max\{Q^*(1, X), Q^*(0, X)\}] = \mathbb{E}[\max\{\tau^*(X), 0\} + Q^*(0, X)]$$

where $Q^*(a, x)$ is the above defined Q-function and τ^* is the CATE. In practice, we do not know Q^* and thus must estimate it from data. Since the above estimating equation is sensitive to errors in the nuisance Q^* , one may want to subtract a correction term involving a new nuisance function to perform first-order de-biasing. However, because the random quantity $\max\{\tau^*(X), 0\}$ is not differentiable when $\tau^*(X) = 0$, if $\mathbb{P}(\tau^*(X) = 0) > 0$, one is not able to directly apply debiasing methods from the semi-parametric statistics/causal inference literature.

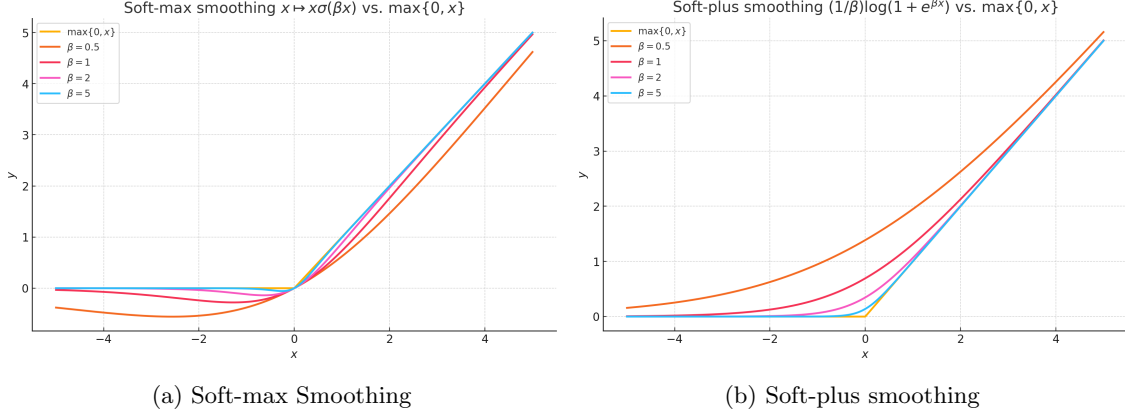


Figure 1: A comparison of softmax smoothing and soft-plus smoothing. Both smoothed approximations closely match the behavior of $\max\{0, x\}$ when $|x|$ is large. However, the softmax mapping $x\sigma(\beta x)$ matches $\max\{0, x\}$ at $x = 0$.

Many existing works that focus on the estimation of irregular parameters, such as Semenova [2023] and Levis et al. [2023], assume that the probability of non-differentiability is zero. However, this may be unrealistic in practice. For instance, in the setting of medical trials, there are likely many individuals who are non-responders, i.e. those for which the medicine has no discernible impact on expected outcome. In fact, a medication could be entirely ineffective altogether. A reasonable approach for estimating a quantity such as the optimal policy value should thus account for this irregularity.

Our approach to handling this non-differentiability, inspired by a similar approach in Chen et al. [2023] in a parametric setting, is to replace the non-differentiable $\max\{0, u\}$ by a smoothed approximation. In particular, letting $\sigma(u) := \frac{e^u}{1+e^u}$ denote the sigmoid function, we consider the softmax approximation

$$\varphi_\beta(\tau(X)) := \tau(X)\sigma(\beta\tau(X)) = \tau(X) \frac{e^{\beta\tau(X)}}{1 + e^{\beta\tau(X)}},$$

where $\tau(x) = Q(1, x) - Q(0, x)$ is an arbitrary CATE estimate. Considering such smoothed approximations is not new in the estimation of irregular parameters. However, by and large, authors have instead opted to consider the soft-plus (or log-sum-exp) approximation $u \mapsto \frac{1}{\beta} \log(1 + e^{\beta u})$ (see Levis et al. [2023], Goldberg et al. [2014]). While this latter smoother matches the behavior of the softmax $u\sigma(\beta u)$ as the magnitude of u grows large, it suffers from a generally non-negligible bias of order $1/\beta$ at zero, a fact illustrated in Figure 1.

More broadly, in the case considered for an arbitrary number of treatments or general irregular functionals, we need to consider the softmax operator generalized to multiple arguments. This is given by the function $\mathbf{sm}^\beta : \mathbb{R}^N \rightarrow \mathbb{R}$ defined as

$$\mathbf{sm}^\beta(u) \equiv \mathbf{sm}^\beta\{u_1, \dots, u_N\} := \frac{\sum_{i=1}^N u_i \exp\{\beta u_i\}}{\sum_{j=1}^N \exp\{\beta u_j\}}.$$

We prove various useful analytical properties of $\mathbf{sm}^\beta$ in Lemma A.1 in Appendix C.

Lastly, we discuss a key density assumption we will make in all of our main theorems. This assumption is the one leveraged in works such as Chen et al. [2023] and Luedtke and Van Der Laan [2016], and allows an arbitrarily large probability of the maximizer being non-unique, while enforcing inverse polynomial density growth in a small neighborhood near but excluding zero. This is illustrated in Figure 2. When $\delta > 1$, this condition becomes a *density decay* or *soft margin* condition, leading to the limiting regime of a hard margin as $\delta \rightarrow \infty$. We note $\delta \leq 1$ is the more theoretically interesting case, as this is when there is a non-negligible probability of the sub-optimality gaps/treatment effects being close to zero.

Assumption 3. *We say a non-negative random variable U satisfies the δ polynomial density assumption if there exists $c, H > 0$ such that the random variable $U\mathbb{1}\{U \in (0, c)\}$ admits a Lebesgue density f satisfying $f(u) \leq Hu^{\delta-1}$.*

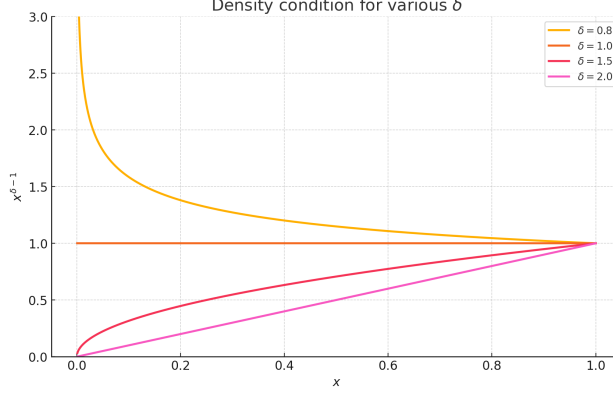


Figure 2: A visualization of the margin condition outlined in Assumption 3 for various choices of $\delta > 0$. For $\delta < 1$, we obtain a density that approaches infinity near zero. For $\delta = 1$, we obtain a bounded density. For $\delta > 1$, we obtain a density that behaves as a polynomial near zero.

3 Inference on the Value of the Optimal Treatment Policy

We now describe how to construct confidence intervals for the value of the optimal treatment policy using observational data. As noted before, the value of the optimal policy is given by

$$V^* = \mathbb{E} \left[\max_{\ell} \{Q^*(\ell, X)\} \right].$$

How one might estimate V^* ? The most naive approach would be to construct an ML estimate $\hat{Q}$ of Q^* on an independent sample and then construct a plug-in estimate: $\hat{V} := \mathbb{P}_n \max_{\ell} \{\hat{Q}(\ell, X)\}$. However, the above equation is “sensitive” to mild perturbations in Q around the true parameter Q^* . That is, the score doesn’t satisfy *Neyman orthogonality*, which is formally defined as follows.

Definition 3.1. Suppose we have a score $m(Z; g)$ that depends on a random variable Z and a nuisance component $g \in L^2(P_W)$, where $g^* \in L^2(P_W)$ denotes the true nuisance function. We say $m(Z; g)$ is *Neyman orthogonal* if, for any $\omega \in L^2(P_W)$, we have

$$D_g \mathbb{E}[m(Z; g^*)](\omega) = 0.$$

Heuristically, a score is Neyman orthogonal if any errors in nuisance estimation are only second order in nature. A standard example of a Neyman orthogonal score is the doubly-robust pseudo-outcome leveraged in the estimation of average treatment effects. Given Definition 3.1, the standard semi-parametric approach to estimating V^* would be to construct a de-biased score. This is usually accomplished by subtracting a correction term that, in expectation, equals the Gateaux derivative $D_Q \mathbb{E} \max_{\ell} \{Q^*(\ell, X)\}(\hat{Q} - Q^*)$. However, the non-differentiability of the maximum function generally prevents us from building such a correction. Instead, using the soft-max function defined in Section 2, we can switch to instead performing inference on a smoothed estimate V^{β} of V^* for some temperature $\beta > 0$:

$$\underbrace{V^* := \mathbb{E} \left[\max_{\ell} \{Q^*(\ell, X)\} \right]}_{\text{Non-differentiable score}} \implies \underbrace{V^{\beta} := \mathbb{E} \left[\mathbf{sm}_{\ell}^{\beta} \{Q^*(\ell, X)\} \right]}_{\text{Smoothed approximation}}.$$

Because the score $\mathbf{sm}_{\ell}^{\beta} \{Q(\ell, x)\}$ is differentiable (in fact, it is C^{∞}) for any fixed $\beta > 0$, we can use first-order debiasing to arrive at a Neyman orthogonal score. The following proposition details precisely what such an orthogonal score would look like. We provide a proof in Appendix B.

Proposition 3.2. For any $\beta > 0$, consider the score Ψ^β defined as

$$\Psi^\beta(z; Q, \alpha) := \mathbf{sm}_\ell^\beta Q(\ell, x) - \sum_{\ell=1}^N \alpha_\ell(a, x)(y - Q(a, x)),$$

where the true nuisances are $Q^*(a, x) = \mathbb{E}[Y \mid X = x, A = a]$ and

$$\alpha_k^\beta(a, x) := \partial_k \mathbf{sm}_\ell^\beta \{Q^*(\ell, x)\} \frac{\mathbb{1}\{a = k\}}{p^*(k \mid x)},$$

where the partial derivative of the softmax function is given by

$$\partial_k \mathbf{sm}_\ell^\beta \{u\} = \frac{e^{\beta u_k}}{\sum_j e^{\beta u_j}} \left[1 - \beta \left\{ u_k - \mathbf{sm}_\ell^\beta u \right\} \right] \quad (3)$$

for any $u \in \mathbb{R}^N$. Then, Ψ^β is Neyman orthogonal and $V^\beta = \mathbb{E} [\Psi^\beta(Z; \mu^*, \alpha^\beta)]$.

In general, the solution to the smoothed, de-biased score V^β is not the same as the optimal policy value V^* . However, by appropriately selecting an increasing sequence of smoothing parameter $(\beta_n)_{n \geq 1}$, one can still ensure that $\sqrt{n}(\hat{V} - V^*)$ is asymptotically normal, where $\hat{V} = \mathbb{P}_n \Psi^{\beta_n}(Z; \hat{Q}_n, \hat{\alpha}_n)$ and $(\hat{Q}_n, \hat{\alpha}_n)$ are nuisance estimates constructed on an independent sample. This is formalized in the following theorem.

Theorem 3.3. Let $(\beta_n)_{n \geq 1}$ be a sequence of smoothing parameters. Let $(Z_n)_{n \geq 1}$ be a sequence of i.i.d. random variables, and let $(\hat{Q}_n)_{n \geq 1}$ and $(\hat{\alpha}_n)_{n \geq 1}$ be sequence of nuisance estimates for Q^* and $\alpha^{\beta_n} = (\alpha_\ell^{\beta_n} : \ell \in [N])$ such that $\hat{Q}_n, \hat{\alpha}_n$ are independent of $Z_1, \dots, Z_n$. Suppose the following assumptions hold in addition to Assumption 1.

1. (Controlled Density Near Zero) We assume that, for each $k \in [N]$, the random variable $\Delta_k := \max_\ell \{Q^*(\ell, x)\} - Q^*(k, x)$ satisfies Assumption 3 with constants $c, H, \delta > 0$.
2. (Growing Smoothing Parameters) We assume the smoothing parameters (β_n) satisfy $\beta_n = \omega \left(n^{\frac{1}{2(1+\delta)}} \right)$.
3. (Nuisance Convergence) We assume the nuisance estimates convergence at sufficiently fast rates, in the particular that

$$\|\hat{Q}_n - Q^*\|_{L^2(P_X)} = o_{\mathbb{P}}(n^{-1/4} \beta_n^{-1/2}) \quad \text{and} \quad \|\hat{\alpha}_{n,\ell} - \alpha_\ell^{\beta_n}\|_{L^2(P_W)} \cdot \|\hat{Q}_n - Q^*\|_{L^2(P_W)} = o_{\mathbb{P}}(n^{-1/2}) \quad \text{for all } \ell \in [N].$$

4. (Boundedness) There is some universal constant $B > 0$ such that:

$$\|Q^*\|_{L^\infty(P_W)}, \|\hat{Q}_n\|_{L^\infty(P_W)}, \|\alpha^{\beta_n}\|_{L^\infty(P_W)}, \|\hat{\alpha}_n\|_{L^\infty(P_W)}, \|Y\|_{L^\infty(P_Y)} \leq B.$$

Then, letting $\hat{V}_n := \mathbb{P}_n \Psi^{\beta_n}(Z; \hat{Q}_n, \hat{\alpha}_n)$, we have the following asymptotic linearity result:

$$\sqrt{n}(\hat{V}_n - V^*) := \frac{1}{\sqrt{n}} \sum_{i=1}^n \{\Psi^*(z; Q^*, \alpha^*) - V^*\} + o_{\mathbb{P}}(1).$$

Consequently, asymptotic normality also holds:

$$\sqrt{n}(\hat{V}_n - V^*) \Rightarrow \mathcal{N}(0, \Sigma^*)$$

where $\Sigma^* := \text{Var}[\Psi^*(Z; Q^*, \alpha^*)]$.

The following corollary allows us to build confidence intervals using the quantiles of the standard normal distribution.

Corollary 3.4. *Under the conditions of Theorem 3.3, letting $\hat{\Sigma}_n := \frac{1}{n} \sum_{m=1}^n \left\{ \Psi^{\beta_n}(Z_m; \hat{Q}_n, \hat{\alpha}_n) - \hat{V}_n \right\}^2$, we have*

$$\sqrt{n} \hat{\Sigma}_n^{-1/2} (\hat{V}_n - V^*) \Rightarrow \mathcal{N}(0, 1).$$

Consequently, for any $\delta \in (0, 1)$, the set

$$C_{1-\delta} := \left[\hat{V}_n - \frac{\hat{\Sigma}_n^{1/2}}{\sqrt{n}} z_{\delta/2}, \hat{V}_n + \frac{\hat{\Sigma}_n^{1/2}}{\sqrt{n}} z_{\delta/2} \right]$$

forms an asymptotic $1 - \delta$ confidence interval for V^* , where z_γ is the γ th quantile of the standard normal distribution.

We now provide a very brief proof sketch illustrating how one should select β_n to ensure asymptotic normality.

Proof Sketch for Theorem 3.3. For any $\beta > 0$, let $\text{Bias}(\beta) := V^\beta - V^*$ denote the bias incurred through smoothing and let $\text{Rem}(\beta)$ be a bound on a second order Taylor expansion remainder of $\mathbb{E} \left[\mathbf{sm}_\ell^\beta \{ \hat{Q}_n(\ell, X) \} \right]$ around $V^\beta = \mathbb{E} \left[\mathbf{sm}_\ell^{\beta_n} \{ Q^*(\ell, X) \} \right]$, given by

$$\text{Rem}(\beta) := \frac{1}{2} D_Q^2 \mathbb{E} \left[\mathbf{sm}_\ell^\beta \{ \bar{Q}(\ell, X) \} \right] (\hat{Q}_n - Q^*)$$

for $\bar{Q} = \bar{t}Q^* + (1 - \bar{t})\hat{Q}_n$ for some $\bar{t} \in [0, 1]$. After some manipulation of terms and under standard $o_{\mathbb{P}}(n^{-1/4})$ nuisance estimation rates, one arrives at the decomposition

$$\begin{aligned} \sqrt{n}(\hat{V} - V^{\beta_n}) &= \sqrt{n}\text{Bias}(\beta_n) + \underbrace{\frac{1}{\sqrt{n}} \sum_{m=1}^n \{ \Psi^{\beta_n}(Z_m; Q^*, \alpha^{\beta_n}) - V^{\beta_n} \}}_{\text{Asymptotically Normal Term}} \\ &\quad + \underbrace{\frac{\sqrt{n}}{2} \sum_{\ell} \|\hat{Q}_n - Q^*\|_{L^2(P_W)} \cdot \|\hat{\alpha}_{n,\ell} - \alpha_\ell^{\beta_n}\|_{L^2(P_W)}}_{=o_{\mathbb{P}}(1) \text{ by assumption}} + \sqrt{n}\text{Rem}(\beta_n) + o_{\mathbb{P}}(1). \end{aligned}$$

The “asymptotically normal term”, as the name would suggest, will converge to a normal distribution via an application of an appropriate central limit theorem for triangular arrays. To obtain overall asymptotic linearity/normality, one just needs to show that $\text{Bias}(\beta_n) = o(n^{-1/2})$ and $\text{Rem}(\beta_n) = o_{\mathbb{P}}(n^{-1/2})$. Leveraging bounds on the second derivative of the softmax function, one can show $\text{Rem}(\beta_n) \lesssim \beta_n \|\hat{Q} - Q^*\|_{L^2(P_W)}^2$. Likewise, if one assumes that the density $f_k(u)$ of the absolute value of random sub-optimality gaps $\Delta_k := \max_{\ell} Q^*(\ell, X) - Q^*(k, \ell)$ is bounded by $u^{\delta-1}$ in a small interval $(0, c)$, then one can show $\text{Bias}(\beta_n) = \left(\frac{1}{\beta_n} \right)^{1+\delta}$. Rearranging, one finds that, so long as $\beta_n = \omega \left(n^{\frac{1}{2(1+\delta)}} \right)$ and $\|\hat{Q} - Q^*\|_{L^2(P_W)} = o_{\mathbb{P}}(n^{-1/4} \beta_n^{-1/2})$, one obtains desired asymptotic normality. $\square$

While the above is just a very heuristic sketch, a formal proof follows as a corollary of Theorem 4.2. We provide the full proof in Appendix B. We now provide some further intuition and interpretation for the result.

Remark 3.5. We start by commenting on the density assumption of Theorem 3.3 in the case of two treatments $A = 0$ and $A = 1$. In this setting, for $a \in \{0, 1\}$, $\Delta_a = 0$ on the event $\{Q^*(a, X) > Q^*(1 - a, X)\}$ and $\Delta_a = |\tau^*(X)|$ otherwise. Thus, if the absolute value of the CATE $|\tau^*(X)|$ satisfies the density condition outlined in Assumption 3, the random sub-optimality gaps Δ_0 and Δ_1 satisfy it with the same parameters c, H , and δ . We again emphasize that this density condition allows the probability of non-response $\mathbb{P}(\tau^*(X) = 0)$ to be arbitrarily large.

Remark 3.6. We briefly spend some time parsing the density assumption in the context of work by Luedtke and Van Der Laan [2016]. First, suppose we were able to exactly take $\beta_n = \Theta\left(n^{\frac{1}{2(1+\delta)}}\right)$. Then, we would need a rate of $\|\hat{Q}_n - Q^*\|_{L^2(P_W)} = o_{\mathbb{P}}\left(n^{-\frac{1}{4}\frac{2+\delta}{1+\delta}}\right)$, which is precisely the rate obtained by the estimator in Luedtke and Van Der Laan [2016]. We consider several different density assumptions on τ^* below.

1. (*Bounded Density*) The case $\delta = 1$ corresponds to the density of the sub-optimality gaps Δ_i being bounded near, but not at zero. In this case, the learner must estimate the Q-functions at a $o_{\mathbb{P}}(n^{-3/8})$ rate in the $L^2(P_W)$ norm. This departs from the traditional $o_{\mathbb{P}}(n^{-1/4})$ rates required on nuisance estimates when the score is twice differentiable and has bounded Hessian. It is still slower than the super-parametric rate of $o_{\mathbb{P}}(n^{-1/2})$ that would be required from non-orthogonal scores.
2. (*Unbounded Density near Zero*) When $\delta = 1/2$, the density of the gaps approaches infinity near zero. In this case, we need a rate of $o_{\mathbb{P}}(n^{-5/12})$ for regression estimates, which is naturally faster than what was required in the setting of a bounded density.
3. (*Quadratic Density near Zero*) As a final example, when $\delta = 2$, the density of $\tau^*(X)$ puts very little mass near the origin. Because of this, a learner can get away with slower $o_{\mathbb{P}}(n^{-1/3})$, which is slower than the case of bounded density. More generally, as $\delta \rightarrow \infty$, the closer we get to being able to assume $o_{\mathbb{P}}(n^{-1/4})$ estimation rates on the regression.

One subtlety in our setting is that we cannot actually take β_n to be on the order of $n^{\frac{1}{2(1+\delta)}}$ — rather we must have $\beta_n = \omega\left(n^{\frac{1}{2(1+\delta)}}\right)$. Thus, we will estimate nuisances at slightly faster rates than those described above. However, we view this as minor, as we can simply take an arbitrarily slowly growing function $f(n)$ (say $f(n) := \log^*(n)$) and set $\beta_n := n^{\frac{1}{2(1+\delta)}} f(n)$. Then, we would need a nuisance estimation rate of $\|\hat{Q} - Q^*\|_{L^2(P_W)} = o_{\mathbb{P}}\left(n^{-\frac{1}{4}\frac{2+\delta}{1+\delta}} f(n)^{-1}\right)$. We note that this same interpretation of necessary nuisance rates holds in Section 4 below as well.

4 Softmax Smoothing for General Irregular Functionals

We now extend the softmax smoothing techniques we have used in estimating the value of the optimal treatment policy to more general classes of irregular functionals. As noted in Section 2, we adopt the general framework considered in Semenova [2023], which defines V^* as the expected point-wise maximum over a class of scores involving nuisance components, i.e.

$$V^* := \mathbb{E} \left[\max_{k=1}^N \psi_k(X; g_k^*) \right].$$

While this framework is quite general, the estimators proposed by Semenova [2023] suffer several key limitations. First, all convergence rates of nuisance estimates must be $o(n^{-1/4})$ in the $L^\infty(P_Z)$ norm, an exceedingly strong requirement for most practical settings.¹ Second, and perhaps more problematically, the argument maximizer is assumed to be *unique almost surely*, e.g. in the case covered in the previous sections the probability of non-response to treatment is assumed to be zero. In this section, we use softmax smoothing to construct confidence intervals (a) only assuming L^2 rates of convergence and (b) allowing for an arbitrarily large probability of the argument maximizer being non-unique.

Once again, the first step in our recipe is to construct a softmax approximation to the main objective. To this end, we define

$$V^\beta := \mathbb{E} \left[\mathbf{sm}_k^\beta \psi_k(X; g_k^*) \right].$$

As before, the quantity $\mathbb{E}[\mathbf{sm}_\ell^\beta \psi_\ell(X; g_\ell)]$ is, without modification, highly sensitive to misestimation in the nuisance components $g_1, \dots, g_N$. Thus, we should subtract a first order correction/de-biasing term to arrive at a Neyman orthogonal score. The following proposition accomplishes precisely this — we provide a proof in Appendix C.

¹As a simple example, let $f_0 : [0, 1] \rightarrow [0, 1]$ be given by $f_0(x) = 0$ and $f_n : [0, 1] \rightarrow [0, 1]$ be $f_n(x) := \mathbb{1}\{x \leq n^{-1/3}\}$. Then, if $\mathbb{P}_X$ is the uniform distribution on $[0, 1]$, we have $\|f_n - f_0\|_{L^2(P_X)} = o_{\mathbb{P}}(n^{-1/4})$ but $\|f_n - f_0\|_{L^\infty(P_X)} = 1$ for all n .

Proposition 4.1. *For any $\beta > 0$, consider the score*

$$\Psi^\beta(z; g, \alpha) := \mathbf{sm}_\ell^\beta \psi_\ell(x; g_\ell) + \sum_{\ell=1}^N \alpha_\ell(w)^\top (u_k - g_k(w))$$

where the k th additional nuisance α_k^β is given by

$$\alpha_k^\beta(w) := \partial_k \mathbf{sm}_\ell^\beta(\psi_\ell(x; g_\ell^*)) \cdot \zeta_k^*(w),$$

Then, Ψ^β is Neyman orthogonal, and we have $V^\beta = \mathbb{E}[\Psi^\beta(Z; g^*, \alpha^\beta)]$, where the explicit formula for $\partial_k \mathbf{sm}_\ell^\beta\{u\}$ is given in Equation (3).

Using the above Neyman orthogonal score, we can arrive at a general asymptotic linearity/normality result that allows one to construct confidence intervals for broad classes of irregular statistical parameters. Of particular importance will be the following, limiting score

$$\Psi^*(z; g, \alpha) = \max_\ell \{\psi_\ell(z; g_\ell^*)\} + \sum_{i=1}^N \alpha(w)^\top (u_k - g_k(w)),$$

where the limiting k th nuisance α_k^* is given as the point-wise limit $\alpha_k^*(w) := \lim_{\beta \rightarrow \infty} \alpha_k^\beta(w)$, i.e. the quantity

$$\alpha_k^*(w) = \mathfrak{s}_k(\psi_\ell(x; g_\ell^*)) \zeta_k^*(w) \quad \mathfrak{s}_k(u) := \frac{\mathbb{1}\{k \in \arg \max_\ell \{u_1, \dots, u_N\}\}}{|\arg \max_\ell \{u_1, \dots, u_N\}|}.$$

This is similar to the score outlined in Equation (1), but is just generalized to a broader class of irregular parameters. We now state our main result, which we prove in Appendix C

Theorem 4.2. *Let $(\beta_n)_{n \geq 1}$ be a sequence of smoothing parameters. Let $(Z_n)_{n \geq 1}$ be a sequence of i.i.d. random variables, and let $(\hat{g}_n)_{n \geq 1}$ and $(\hat{\alpha}_n)_{n \geq 1}$ be a sequence of nuisance estimates for $g^* := (g_\ell^* : \ell \in [N])$ and $\alpha^{\beta_n} := (\alpha_\ell^{\beta_n} : \ell \in [N])$, such that $\hat{g}_n$ and $\hat{\alpha}_n$ are independent of $Z_1, \dots, Z_n$. Suppose the following assumptions hold.*

1. (Controlled Density Near Zero) Let $M := \max_\ell \psi_\ell(X; g_\ell^*)$ and for each ℓ let $\Delta_\ell := M - \psi_\ell(X; g_\ell^*)$. We assume for each ℓ that Δ_ℓ satisfies Assumption 3 with constants $c, H, \delta > 0$.
2. (Growing Smoothing Parameters) We assume $(\beta_n)_{n \geq 1}$ are chosen to satisfy $\beta_n = \omega\left(n^{\frac{1}{2(1+\delta)}}\right)$.
3. (Mean Squared Continuity) Each constituent score $\psi_k(Z; g_k)$ satisfies mean-squared continuity, i.e. we have

$$\|\psi_k(X; g_k) - \psi_k(X; g'_k)\|_{L^2(P_X)} \lesssim \|g_k - g'_k\|_{L^2(P_W)}^\gamma \quad \text{for some } \gamma > 0.$$

4. (Nuisance Rates) We have for each $k \in [N]$ that

$$\|\hat{g}_{n,k} - g_k^*\|_{L^2(P_W)} = o_{\mathbb{P}}(n^{-1/4} \beta_n^{-1/2}) \quad \text{and} \quad \|\hat{\alpha}_{n,k} - \alpha_k^{\beta_n}\|_{L^2(P_W)} \cdot \|\hat{g}_{n,k} - g_k^*\|_{L^2(P_W)} = o_{\mathbb{P}}(n^{-1/2}).$$

5. (Boundedness) There is a constant $B > 0$ such that

$$\|g^*\|_{L^\infty(P_W)}, \|\hat{g}_n\|_{L^\infty(P_W)}, \|\alpha^{\beta_n}\|_{L^\infty(P_W)}, \|\hat{\alpha}_n\|_{L^\infty(P_W)}, \|U_k\|_{L^\infty(P_Z)} \leq B.$$

6. (Score Derivative Regularity) We assume with probability 1 that

$$\mathbb{E}_X [D_{g_k} \psi_k(X; \bar{g}_k) (\hat{g}_{n,k} - g_k^*)^2], \mathbb{E}_X [D_{g_k}^2 \psi_k(X; \bar{g}_k) (\hat{g}_{n,k} - g_k^*)] \lesssim \|\hat{g}_{n,k} - g_k^*\|_{L^2(P_W)}^2$$

for all $\bar{g} \in [g^*, \hat{g}_n]$. Further, letting $F_{\nu, C}$ be as in Assumption 2, we assume for any $C > 0$ that $\mathbb{E} F_{\hat{g}_n - g^*, C} < \infty$ for all $n \geq 1$.

Then, letting $\widehat{V}_n := \mathbb{P}_n \Psi^{\beta_n}(Z; \widehat{g}_n, \widehat{\alpha}_n)$, we have the asymptotic linearity result

$$\sqrt{n}(\widehat{V}_n - V^*) = \frac{1}{\sqrt{n}} \sum_{m=1}^n \{\Psi^*(Z_m; g^*, \alpha^*) - V^*\} + o_{\mathbb{P}}(1),$$

where Ψ^* and α^* are as defined above. Consequently, we have

$$\sqrt{n}(\widehat{V}_n - V^*) \Rightarrow \mathcal{N}(0, \Sigma^*),$$

where $\Sigma^* := \text{Var}[\Psi^*(Z; g^*, \alpha^*)]$.

As before, the above theorem yields a corollary that allows for the practical construction of asymptotically valid confidence intervals using a plug-in estimate of the variance.

Corollary 4.3. *Under the conditions of Theorem 4.2, letting $\widehat{\Sigma}_n := \frac{1}{n} \sum_{m=1}^n \left\{ \Psi^{\beta_n}(Z_m; \widehat{g}_n, \widehat{\alpha}_n) - \widehat{V}_n \right\}^2$, we have*

$$\sqrt{n\widehat{\Sigma}_n^{-1/2}}(\widehat{V}_n - V^*) \Rightarrow \mathcal{N}(0, 1).$$

Consequently, for any $\delta \in (0, 1)$, the set

$$C_{1-\delta} := \left[\widehat{V}_n - \frac{\widehat{\Sigma}_n^{1/2}}{\sqrt{n}} z_{\delta/2}, \widehat{V}_n + \frac{\widehat{\Sigma}_n^{1/2}}{\sqrt{n}} z_{\delta/2} \right]$$

forms an asymptotic $1 - \delta$ confidence interval for V^* , where z_γ is the γ th quantile of the standard normal distribution.

Remark 4.4. In the statements of both of our main theorems (Theorem 3.3 and Theorem 4.2), we have assumed that the nuisance estimates are constructed on an independent sample of data. In practice, one will typically want to use the cross-fitting algorithm of Chernozhukov et al. [2018] to produce the estimate $\widehat{V}_n$, as this will make the most efficient use of data. In short, cross-fitting works as follows. Let K be fixed number of folds.

1. Given a sample $Z_1, \dots, Z_n$, splits the indices $[n]$ into K disjoint sets (or “folds”) $\mathcal{I}_1, \dots, \mathcal{I}_K$ of size n/K . One can canonically set $\mathcal{I}_k := \{m \leq n : m = ck \text{ for some } c\}$.
2. Letting $\mathcal{I}_{-k} := [n] \setminus \mathcal{I}_k$, we fit nuisance estimates $\widehat{g}_n^{(-k)}, \widehat{\alpha}_n^{(-k)}$ using ML methods on samples with indices in $\mathcal{I}_{-k}$.
3. We return the estimate $\widehat{V}_n$ given by

$$\widehat{V}_n := \frac{1}{n} \sum_{k=1}^K \sum_{i \in \mathcal{I}_k} \Psi^{\beta_n}(Z_i; \widehat{g}_n^{(-k)}, \widehat{\alpha}_n^{(-k)}).$$

If, for each $k \in [K]$, the nuisance estimates $\widehat{g}_n^{(-k)}$ and $\widehat{\alpha}_n^{(-k)}$ satisfy the assumptions of Theorem 4.2, then it is simple to show that

$$\sqrt{n}(\widehat{V}_n - V^*) = \frac{1}{\sqrt{n}} \sum_{m=1}^n \{\Psi^*(Z_m; g^*, \alpha^*) - V^*\} + o_{\mathbb{P}}(1).$$

Further, one will have

$$\sqrt{n\widehat{\Sigma}_n^{-1/2}}(\widehat{V}_n - V^*) \Rightarrow \mathcal{N}(0, 1) \quad \text{where} \quad \widehat{\Sigma}_n := \frac{1}{n} \sum_{k=1}^K \sum_{i \in \mathcal{I}_k} \left\{ \Psi^{\beta_n}(Z_i; \widehat{g}_n^{(-k)}, \widehat{\alpha}_n^{(-k)}) - \widehat{V}_n \right\}^2,$$

a fact with which one can construct confidence intervals in the same manner as outlined in Corollary 4.3. We provide a brief proof in Appendix C in Corollary C.2.

4.1 An Application to Conditional Balke and Pearl Bounds

Throughout this work, we have been using the value of the optimal treatment policy $V^* := \mathbb{E}[\max_{\ell} \{Q^*(\ell, X)\}]$ as the quintessential example of an irregular causal parameter. We now discuss another, practically relevant example focused on the partial identification of the average treatment effect (ATE) in instrumental variable settings. We largely follow the exposition given in Levis et al. [2023], and recommend Balke and Pearl [1994], Balke [1995], Balke and Pearl [1997] for further exposition and history. In general, one can consider both upper and lower bounds — we solely discuss lower bounds as results for upper bounds are exactly analogous. The main contributions of this work in comparison to the results of Levis et al. [2023] are that (a) our main theorem allows one to select $(\beta_n)_{n \geq 1}$ to guarantee asymptotic normality around the unsmoothed, population Balke and Pearl bounds (defined below) and (b) the theorem holds even when the argument maximizer used in defining the Balke and Pearl bounds is non-unique with arbitrarily large probability. We discuss both of these points in further detail below.

For this example, observations are of the form $Z = (X, A, V, Y)$, $X \in \mathcal{X}$ represent covariates, $A \in \{0, 1\}$ represents a binary treatment, $V \in \{0, 1\}$ a binary instrument, and $Y \in \{0, 1\}$ a binary outcome. For simplicity, we let $W = (X, V)$ denote the tuple consisting of covariates and instrument. We make the following assumptions on the vector of observations Z .

Assumption 4. *We assume the learner observes $Z = (X, A, V, Y)$ as outlined above and that the following conditions hold.*

1. *(Consistency) There exist potential outcomes $A(z)$ and $Y(a, z)$ for $a, z \in \{0, 1\}$ such that $A = A(Z)$ and $Y = Y(A, Z)$.*
2. *(Exclusion Restriction) The potential outcomes only depend on v through a , i.e. we have $Y(0, a) = Y(1, a)$.*
3. *(Conditional Ignorability) The instrument is conditionally independent of the treatment and outcome given covariates, i.e. $V \perp\!\!\!\perp A(v), Y(v, A(v)) \mid X$.*
4. *(Strong Positivity) Let $p^*(v \mid x) := \mathbb{P}(V = v \mid X = x)$. We assume $\eta \leq p^*(1 \mid X) \leq 1 - \eta$ almost surely for some $\eta \in (0, 1/2]$.*

The goal is to provide a lower bound on the average treatment effect $\mathbb{E}[Y(A = 1) - Y(A = 0)] = \mathbb{E}[Y(V, 1) - Y(V, 0)]$, which in general is not point identified. We will show how to fit this example into the framework of this section. Then, we will compare our results to those of Levis et al. [2023].

For any $a, y \in \{0, 1\}$, define the nuisance function $q_{ya}^*(x, v) := \mathbb{P}(Y = y, A = a \mid X = x, V = v)$. Next, define the scores $\psi_1(x; q_1), \dots, \psi_8(x; q_8)$ as follows:

$$\begin{aligned} \psi_1(x; q_1) &= q_{11}(x, 1) + q_{00}(x, 0) - 1 & \psi_2(x; q_2) &= q_{11}(x, 0) + q_{00}(x, 1) - 1 \\ \psi_3(x; q_3) &= -q_{01}(x, 1) - q_{10}(x, 1) & \psi_4(x; q_4) &= -q_{01}(x, 0) - q_{10}(x, 0) \\ \psi_5(x; q_5) &= q_{11}(x, 0) - q_{11}(x, 1) - q_{10}(x, 1) - q_{01}(x, 0) - q_{10}(x, 0) \\ \psi_6(x; q_6) &= q_{11}(x, 1) - q_{11}(x, 0) - q_{10}(x, 0) - q_{01}(x, 1) - q_{10}(x, 1) \\ \psi_7(x; q_7) &= q_{00}(x, 1) - q_{01}(x, 1) - q_{10}(x, 1) - q_{01}(x, 0) - q_{00}(x, 0) \\ \psi_8(x; q_8) &= q_{00}(x, 0) - q_{01}(x, 0) - q_{10}(x, 0) - q_{01}(x, 1) - q_{00}(x, 1). \end{aligned}$$

For each $k \in [8]$, let $\mathcal{I}_k \subset \{0, 1\}^2$ denote the set of pairs (y, a) such that $q_{ya}^*(x, v)$ is present in the equation $\psi_k(x; q_\ell)$ and let $d_k := |\mathcal{I}_k|$ denote the number of tuples present in the k th equation. The true, unknown nuisance for ψ_k is $q_k^* := (q_{ya}^* : (y, a) \in \mathcal{I}_k)$. Levis et al. [2023] show that one has the following lower bound on the ATE:

$$V^* := \mathbb{E}[\max_{\ell=1}^8 \psi_\ell(X; q_\ell^*)] \leq \mathbb{E}[Y(A = 1) - Y(A = 0)].$$

Our goal is to show that our smoothing approach can be used to perform inference on the lower Balke and Pearl bound V^* . First, we show that the constituent scores satisfy Assumption 2. Observe that, for any pair (y, a) , we have the identity

$$q_{ya}^*(x, v) = \mathbb{E}[\mathbb{1}\{Y = y, A = a\} \mid X = x, V = v].$$

Thus, letting $U_{ya} := \mathbb{1}\{Y = y, A = a\}$ and $U_\ell := (U_{ya} : (y, a) \in \mathcal{I}_\ell)$, it is clear that we have $q_\ell^*(x, v) = \mathbb{E}[U_\ell \mid X = x, V = v]$. Thus the first part of Assumption 2 is satisfied. The second part of Assumption 2 is also satisfied by noting that, for any $(y, a, v) \in \{0, 1\}^3$ and $\omega \in L^2(P_W)$, we have $D_q q_{ya}(X, v)(\omega) = \mathbb{E}[\zeta_{ya}(X, V)\omega(X, V) \mid X]$, where $\zeta_{ya}(X, V) := \frac{\mathbb{1}\{V=v\}}{p^*(v|X)}$, so the Riesz representer for score k is the vector-valued function given by

$$\zeta_k^*(x, v) = (\text{sgn}_k(y, a, v) \cdot \zeta_{ya}(x, v) : (y, a) \in \mathcal{I}_k),$$

where $\text{sgn}_k(y, a, v) = 1$ if $q_{ya}(\cdot, v)$ appears in score ψ_k with positive sign, $\text{sgn}_k(y, a, v) = -1$ if it appears with negative sign, and $\text{sgn}_k(y, a, v) = 0$ otherwise. To see that the third assumption is satisfied, note that for any x and $\nu : \mathcal{W} \rightarrow \mathbb{R}^d$ (where $d = d_1 + \dots + d_k$) we have that

$$\begin{aligned} \ell_{x,\nu}(\epsilon)_k &= \sum_{(y,a,v)} \text{sgn}_k(y, a, v)(q_{ya}^* + \epsilon \nu_{ya})(x, v) - \mathbb{1}\{k \in \{1, 2\}\} \\ \partial \ell_{x,\nu}(\epsilon)_k &= \sum_{(y,a,v)} \text{sgn}_k(y, a, v) \nu_{y,a}(x, v) \\ \partial_\epsilon^2 \ell_{x,\nu}(\epsilon)_k &= 0, \end{aligned}$$

so the zeroth through second directional derivatives are clearly continuous in ϵ , with the second derivative vanishing entirely. Likewise, if $\nu \in L^2(\mathbb{R}^d; P_W)$, we simply take the dominating random variable to be

$$F_{\nu,C} := \left(\sum_{k=1}^8 \left\{ \sum_{(y,a,v)} \text{sgn}_k(y, a, v) \nu_{y,a}(X, v) \right\}^2 \right)^{1/2},$$

which is integrable by Jensen's inequality, the parallelogram inequality, and strong positivity.

To apply Theorem 4.2, one would just need to check that each of the assumptions of the theorem is satisfied — we discuss this in Subsection C.1 of the appendix. We now interpret our results in this setting. First, we note our use of softmax smoothing has several advantages over the soft-plus (or log-sum-exp) smoothing used in Levis et al. [2023]. First, as emphasized throughout this work, we allow an arbitrarily large probability of there being a non-unique argument maximizer, i.e. the set $\arg \max_\ell \{\psi_\ell(x; q_\ell^*)\}$ need not be a singleton almost surely. We note that the bias of soft-plus at zero is on the order $\frac{1}{\beta}$. Since the second order nuisance remainders contain terms of the form $\beta \|\hat{q}_{\ell,n} - q_\ell^*\|_{L^2(P_W)}^2$, one would simultaneously need $\beta_n = \omega(n^{-1/2})$ and $\beta_n = o(n^{-1/2})$ to obtain asymptotic normality, which is impossible.

Second, our smoothing result yields asymptotic normality around the true, un-smoothed lower Balke and Pearl bound V^* . In their work, Levis et al. [2023] do not show how to select smoothing parameters β_n as to obtain asymptotic normality around V^* . While Remark 1 of their work notes that, in a simplified setting, one can show that the bias grows as $O((1/\beta)^{1+\delta})$ instead of $O(1/\beta)$ when a density condition is satisfied, it is not precisely clear what this simplified setting is. Additionally, the authors do not describe how to trade off the bias and second order remainders to achieve normality around the population bound. Further, the density condition considered in Levis et al. [2023] does not allow the argument maximizer to be non-unique. We also emphasize that, in the case of a regular law (i.e. one in which there is probability one of the argument maximizer being unique) the asymptotic score Ψ^* provided above exactly matches the efficient influence function described in section 4.1 of Levis et al. [2023].

5 Conclusion

In this paper, we considered the problem of constructing confidence intervals for the value of the optimal treatment policy. Our approach is intuitive and computationally lightweight. We allow for an arbitrarily large probability of non-response to a treatment in the population, and only require a mild condition on the density of the CATE in a small neighborhood near zero. Our approach can extend to other commonly-encountered examples of irregular parameters, such as Balke and Pearl bounds on the average treatment effect. In sum, we show that softmax smoothing is an effective tool in the estimation of irregular statistical parameters, and perhaps should be considered more widely in the causal inference literature.

While the results presented in this paper are quite general, there are many important open questions that remain to be answered. First, throughout our work, we assume that the sequence of smoothing parameters (β_n) is fixed and chosen according to the density of CATE (more generally, scores) in a small neighborhood of zero. In practice, one may not know this behavior, and thus it would be of great theoretical and practical interest to develop data-adaptive methods for picking the β_n . Second, while we are able to argue efficiency in the setting where the law of the CATE is regular (i.e. $\mathbb{P}(\tau^*(X) = 0) = 0$), it would be interesting to derive non-parametric and semi-parametric lower bounds for irregular laws. To the best of our knowledge, there do not currently exist techniques for doing so.

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A Properties of the Softmax Operator

We now prove a variety of properties for the softmax operator $\mathbf{sm}^\beta$, which we recall, for a vector $u = (u_1, \dots, u_N) \in \mathbb{R}^N$, is defined as

$$\mathbf{sm}_\ell^\beta(u) := \frac{\sum_{i=1}^N u_i \exp\{\beta u_i\}}{\sum_{j=1}^N \exp\{\beta u_j\}}.$$

We start by proving a number of basic analytical properties of the softmax function that will be useful throughout our work. In the sequel, for simplicity, we define the i th softmax weight $w_i^\beta(u)$ to be

$$w_i^\beta(u) := \frac{e^{\beta u_i}}{\sum_j e^{\beta u_j}}$$

Lemma A.1. *For any $\beta > 0$, consider the softmax function $\mathbf{sm}^\beta$ as defined above. The following hold.*

1. *The i th partial derivative of $\mathbf{sm}^\beta(u)$, which is given by*

$$(\nabla_u \mathbf{sm}_\ell^\beta(u))_i = w_i^\beta(u) \left[1 + \beta(u_i - \mathbf{sm}_\ell^\beta(u)) \right],$$

satisfies $\sup_u \|\nabla_u \mathbf{sm}^\beta(u)\|_\infty \leq 1 + \log(N)$.

2. *$\mathbf{sm}^\beta(u)$ is L -Lipschitz with $L := \sqrt{N}(1 + \log(N))$.*
3. *The Hessian of $\mathbf{sm}^\beta(u)$ satisfies $\|\nabla_u^2 \mathbf{sm}^\beta(u)\|_{op} \leq 6\beta + 4\beta \log(N)$.*
4. *The softmax function obeys the following limiting properties:*

$$\mathbf{sm}_\ell^\beta(u_\ell) \xrightarrow{\beta \rightarrow \infty} \max_\ell \{u_\ell\}.$$

5. *The gradient of the softmax function obeys the following limiting properties:*

$$(\nabla_u \mathbf{sm}_\ell^\beta(u))_k \xrightarrow{\beta \rightarrow \infty} \mathfrak{s}_k(u),$$

$$\text{where } \mathfrak{s}_k(u) := \frac{\mathbb{1}\{k \in \arg \max_\ell (u_1, \dots, u_N)\}}{|\arg \max_\ell (u_1, \dots, u_N)|}.$$

Proof. We prove each of the above points in succession.

Proof of Point 1 We start by proving the first point. It suffices to compute the k th partial derivative. Let $f(u) := \sum_i u_i \exp\{\beta u_i\}$ and $g(u) := \sum_j \exp\{\beta u_j\}$. Computing the k th partial for each yields.

$$\begin{aligned}\partial_k f(u) &= e^{\beta u_k} + \beta u_k e^{\beta u_k} \\ \partial_k g(u) &= \beta e^{\beta u_k}.\end{aligned}$$

Now, we can compute the overall partial derivative via the quotient rule:

$$\begin{aligned}\partial_k \frac{\sum_i u_i e^{\beta u_i}}{\sum_j e^{\beta u_j}} &= \frac{g(u) \partial_k f(u) - f(u) \partial_k g(u)}{g(u)^2} \\ &= \frac{e^{\beta u_k} (1 + \beta u_k) \sum_j e^{\beta u_j} - \beta e^{\beta u_k} \sum_i u_i e^{\beta u_i}}{\left(\sum_j e^{\beta u_j}\right)^2} \\ &= \frac{e^{\beta u_k}}{\sum_j e^{\beta u_j}} [(1 + \beta u_k) - \beta \mathbf{sm}^\beta(u)] \\ &= w_k^\beta(u) [1 + \beta(u_k - \mathbf{sm}^\beta(u))],\end{aligned}$$

which proves the desired formula for the gradient of the softmax. Now, we can bound the magnitude of any individual coordinate of the gradient vector. We consider two cases. First, suppose $u_k \leq \mathbf{sm}^\beta(u)$. Then, we can bound $|w_k^\beta(u) \beta(u_k - \mathbf{sm}^\beta(u))|$ as follows. Let $\tilde{\Delta}_i := \mathbf{sm}^\beta(u) - u_i$ for $i \in [N]$. We have

$$\begin{aligned}|w_k^\beta(u) \beta(u_k - \mathbf{sm}^\beta(u))| &= \beta \tilde{\Delta}_k \frac{e^{\beta u_k}}{\sum_j e^{\beta u_j}} \\ &= \beta \tilde{\Delta}_k \frac{e^{-\beta \mathbf{sm}^\beta(u)}}{e^{-\beta \mathbf{sm}^\beta(u)}} \frac{e^{\beta u_k}}{\sum_j e^{\beta u_j}} \\ &= \beta \tilde{\Delta}_k \frac{e^{-\beta \tilde{\Delta}_k}}{\sum_j e^{-\beta \tilde{\Delta}_j}} \\ &\leq \sup_{u \geq 0} u e^{-u} = \frac{1}{e}.\end{aligned}$$

In the above, the final inequality follows from the fact that $\sum_j e^{-\beta \tilde{\Delta}_j} \geq 1$, since there is always some index $i^* \in [N]$ such that $\tilde{\Delta}_{i^*} = \mathbf{sm}_\ell^\beta(u) - u_i \leq 0$ and hence $e^{-\beta \tilde{\Delta}_{i^*}} \geq 1$. The final equality follows from the fact that the function $u \mapsto u e^{-u}$ attains its maximum of $\frac{1}{e}$ when $u = 1$. Thus, we can bound the k th partial in this setting by

$$|\partial_k \mathbf{sm}^\beta(u)| \leq w_k^\beta(u) + w_k^\beta(u) |\beta \tilde{\Delta}_k| \leq 1 + \frac{1}{e}.$$

Next, we consider the case where $u_k \geq \mathbf{sm}^\beta(u)$. Here, we have the bound

$$|w_k^\beta(u) [1 + \beta(u_k - \mathbf{sm}^\beta(u))]| \leq 1 + \beta(M - \mathbf{sm}^\beta(u)),$$

where $M := \max_i u_i$ for convenience. Using a standard softmax trick, we have the bound:

$$\begin{aligned}M - \mathbf{sm}^\beta(u) &= \sum_j w_j^\beta(u) (M - u_j) \\ &\leq \frac{1}{\beta} \log \left(\sum_j w_j^\beta(u) e^{\beta(M - u_j)} \right) \\ &= \frac{1}{\beta} \log \left(\frac{e^{\beta M}}{\sum_j e^{\beta u_j}} \sum_j 1 \right) \\ &\leq \frac{1}{\beta} \log(N).\end{aligned}$$

In the above, the first inequality follows from the fact that we have by Jensen's inequality, for any random variables X and parameter $\lambda > 0$,

$$\mathbb{E}[X] \leq \frac{1}{\lambda} \log \mathbb{E} \exp\{\lambda X\}.$$

Plugging this into the bound obtained in the previous display yields:

$$w_k^\beta(u) |1 + \beta(u_k - \mathbf{sm}^\beta(u))| \leq 1 + \log(N).$$

Since we have shown the component-wise upper bound in all cases and since $e^{-1} < \log(N)$ when $N \geq 2$, we are done.

Proof of Point 2 The proof of Lipschitz continuity is straightforward. In particular, let $u, u' \in \mathbb{R}^N$. We have via the mean value theorem that for some $\bar{u} \in [u, u']$ that

$$\begin{aligned} |\mathbf{sm}_\ell^\beta(u) - \mathbf{sm}_\ell^\beta(u')| &= |\langle \nabla_u \mathbf{sm}_\ell^\beta(\bar{u}), u - u' \rangle| \\ &\leq \sup_{\bar{u}} \|\nabla_u \mathbf{sm}_\ell^\beta(\bar{u})\|_2 \|u - u'\|_2 \\ &\leq \sqrt{N} \sup_{\bar{u}} \|\nabla_u \mathbf{sm}_\ell^\beta(u)\|_\infty \|u - u'\|_2 \\ &\leq \sqrt{N} (1 + \log(N)) \|u - u'\|_2, \end{aligned}$$

which proves the desired result.

Proof of Point 3 We now go about bounding the Hessian. First, we compute the (k, ℓ) th coordinate of $\nabla_u^2 \mathbf{sm}^\beta(u)$. Computation similar to that performed in the previous paragraphs yields

$$\partial_{k,\ell} \mathbf{sm}^\beta(u) = \beta w_k^\beta(u) \left[(\delta_{k,\ell} - w_\ell^\beta(u))(1 + \beta(u_k - \mathbf{sm}^\beta(u))) + (\delta_{k,\ell} - w_\ell^\beta(u)) - \beta w_\ell^\beta(u)(u_\ell - \mathbf{sm}^\beta(u)) \right].$$

We now bound the ℓ_1 -norm of any row (equivalently, any column) of the Hessian. We have for any k ,

$$\begin{aligned} \|(\nabla_u^2 \mathbf{sm}^\beta(u))_k\|_1 &\leq \underbrace{\beta w_k^\beta(u) |1 + \beta(u_k - \mathbf{sm}^\beta(u))| \sum_\ell |\delta_{\ell,k} - w_\ell^\beta(u)|}_{T_1} \\ &\quad + \underbrace{\beta w_k^\beta(u) \sum_\ell w_\ell^\beta(u) \beta |u_\ell - \mathbf{sm}^\beta(u)|}_{T_2} \\ &\quad + \underbrace{\beta w_k^\beta(u) \sum_\ell |\delta_{k,\ell} - w_\ell^\beta(u)|}_{T_3}. \end{aligned}$$

Our argument used in the proof of the previous part shows that $w_k^\beta(u) |1 + \beta(u_k - \mathbf{sm}^\beta(u))| \leq 1 + \log(N)$, and so we have

$$T_1 \leq \beta(1 + \log(N)) \sum_\ell |\delta_{\ell,k} - w_\ell^\beta(u)| \leq 2\beta(1 + \log(N)).$$

Next we bound T_2 . First, note that we have $\beta u_\ell = \log w_\ell^\beta(u) + \log \left(\sum_j e^{\beta u_j} \right)$ and $\beta \mathbf{sm}_\ell^\beta(u) = \sum_j w_j^\beta(u) \log w_j^\beta(u) + \log \left(\sum_j e^{\beta u_j} \right)$. Thus we have the bound

$$\beta |u_\ell - \mathbf{sm}_\ell^\beta(u)| = \left| \log w_\ell^\beta(u) - \sum_j w_j^\beta(u) \log w_j^\beta(u) \right| = \left| \log w_\ell^\beta(u) + H(u; \beta) \right|$$

where we have defined $H(u; \beta) := -\sum_j w_j^\beta(u) \log w_j^\beta(u)$ as the entropy of the softmax weights. Consequently, we have

$$\begin{aligned}
T_2 &= \beta w_k^\beta(u) \sum_\ell w_\ell^\beta(u) \beta |u_\ell - \mathbf{sm}^\beta(u)| \\
&= \beta w_k^\beta(u) \sum_\ell w_\ell^\beta(u) \left| \log w_\ell^\beta(u) + H(u; \beta) \right| \\
&\leq \beta w_k^\beta(u) \sum_\ell w_\ell^\beta(u) \left| \log w_\ell^\beta(u) \right| + \beta w_k^\beta(u) \sum_\ell w_\ell^\beta(u) H(u; \beta) \\
&= 2\beta w_k^\beta(u) H(u; \beta) \leq 2\beta \log(N)
\end{aligned}$$

where the final equality follows from $|w_k^\beta(u)| \leq 1$ and $0 \leq H(u; \beta) \leq \log(N)$ since the weights $w_1^\beta(u), \dots, w_N^\beta(u)$ describe a distribution on N elements.

Now, we lastly bound T_3 . In particular we have

$$T_3 = \beta w_k^\beta(u) \sum_\ell |\delta_{k,\ell} - w_\ell^\beta(u)| \leq 2\beta.$$

Putting these together yields a bound of $\|(\nabla_u^2 \mathbf{sm}^\beta(u))_k\|_1 \leq 6\beta + 4\beta \log(N)$. We conclude by noting that

$$\|\nabla_u^2 \mathbf{sm}^\beta(u)\|_{op} \leq \max_k \|(\nabla_u^2 \mathbf{sm}^\beta(u))_k\|_1 \leq 6\beta + 4\beta \log(N).$$

In the above, the first inequality follows from Corollary 6.1.5 of Horn and Johnson [2012], which states that for any complex-valued square matrix $A \in \mathbb{C}^{n \times n}$, one has

$$\|A\|_{op} \leq \min \left\{ \max_i \sum_{j=1}^n |a_{i,j}|, \max_j \sum_{i=1}^n |a_{i,j}| \right\},$$

where $|z| = z\bar{z}$ for any $z \in \mathbb{C}$. This proves the desired result.

Proof of Point 4 The proof here is standard, but we include it for completeness. Letting again $M := \max_i u_i$, $\Delta_i := M - u_i$, and now defining $A := |\arg \max\{u_1, \dots, u_N\}|$, we have

$$\begin{aligned}
\lim_{\beta \rightarrow \infty} \mathbf{sm}_\ell^\beta(u) &= \lim_{\beta \rightarrow \infty} \frac{\sum_i u_i \exp\{\beta u_i\}}{\sum_j \exp\{\beta u_j\}} \\
&= \lim_{\beta \rightarrow \infty} \frac{\sum_i u_i \exp\{-\beta \Delta_i\}}{\sum_j \exp\{-\beta \Delta_j\}} \\
&= \lim_{\beta \rightarrow \infty} \frac{\sum_{i:\Delta_i=0} u_i \exp\{-\beta \Delta_i\}}{\sum_j \exp\{-\beta \Delta_j\}} + \lim_{\beta \rightarrow \infty} \frac{\sum_{i:\Delta_i>0} u_i \exp\{-\beta \Delta_i\}}{\sum_j \exp\{-\beta \Delta_j\}} \\
&= M \lim_{\beta \rightarrow \infty} \frac{A}{A + \sum_{j:\Delta_j>0} \exp\{-\beta \Delta_j\}} + \lim_{\beta \rightarrow \infty} \frac{\sum_{i:\Delta_i>0} u_i \exp\{-\beta \Delta_i\}}{A + \sum_{j:\Delta_j>0} \exp\{-\beta \Delta_j\}} \\
&= M.
\end{aligned}$$

Proof of Point 5 Recall that we have the identity

$$(\nabla_u \mathbf{sm}^\beta(u))_i = \partial_i \mathbf{sm}^\beta(u) = \frac{e^{\beta u_i}}{\sum_j e^{\beta u_j}} (1 + \beta(u_i - \mathbf{sm}^\beta(u))).$$

We consider two cases. First, assume that $i \in \arg \max\{u_1, \dots, u_N\}$. We start by showing that

$$\beta(u_i - \mathbf{sm}^\beta(u)) \xrightarrow{\beta \rightarrow \infty} 0.$$

To see this, let $\mathcal{A} := \arg \max\{u_1, \dots, u_N\}$ and $A = |\mathcal{A}|$. Observe that we have

$$\begin{aligned}
\lim_{\beta \rightarrow \infty} \beta(u_i - \mathbf{sm}^\beta(u)) &= \frac{\sum_j \beta(u_i - u_j) e^{\beta u_j}}{\sum_j e^{\beta u_j}} \\
&= \lim_{\beta \rightarrow \infty} \frac{\sum_{j \notin \mathcal{A}} \beta(u_i - u_j) e^{\beta u_j}}{\sum_{j \in \mathcal{A}} e^{\beta u_j} + \sum_{j \notin \mathcal{A}} e^{\beta u_j}} \\
&= \lim_{\beta \rightarrow \infty} \frac{\sum_{j \notin \mathcal{A}} \beta(u_i - u_j) e^{\beta(u_i - u_j)}}{\sum_{j \in \mathcal{A}} e^{\beta(u_i - u_j)} + \sum_{j \notin \mathcal{A}} e^{\beta(u_i - u_j)}} \\
&= \lim_{\beta \rightarrow \infty} \frac{\sum_{j \notin \mathcal{A}} \beta \Delta_j e^{-\beta \Delta_j}}{A + \sum_{j \notin \mathcal{A}} e^{-\beta \Delta_j}} \\
&= \frac{\lim_{\beta \rightarrow \infty} \sum_{j \notin \mathcal{A}} \beta \Delta_j e^{-\beta \Delta_j}}{A + \lim_{\beta \rightarrow \infty} \sum_{j \notin \mathcal{A}} e^{-\beta \Delta_j}} \\
&= 0
\end{aligned}$$

where the final equality follows from the fact that $\lim_{u \rightarrow \infty} u e^{-u} = 0$ and $\lim_{u \rightarrow \infty} e^{-u} = 0$. Further, using a similar argument, we see that we have

$$\begin{aligned}
\frac{e^{\beta u_i}}{\sum_j e^{\beta u_j}} &= \frac{e^{-\beta \Delta_i}}{\sum_j e^{\beta \Delta_j}} \\
&= \frac{1}{A + \sum_{j \notin \mathcal{A}} e^{-\beta \Delta_j}} \\
&\xrightarrow{\beta \rightarrow \infty} \frac{1}{A}.
\end{aligned}$$

Consequently, when $i \in \mathcal{A}$, we have

$$\lim_{\beta \rightarrow \infty} \frac{e^{\beta u_i}}{\sum_j e^{\beta u_j}} \left[1 + \beta(u - \mathbf{sm}_\ell^\beta u) \right] = \frac{1}{A}.$$

Now, we consider the case where $i \notin \mathcal{A}$. We already know that $\frac{e^{\beta u_i}}{\sum_j e^{\beta u_j}} \rightarrow 0$ as $\beta \rightarrow \infty$, and further we see we have

$$\begin{aligned}
\lim_{\beta \rightarrow \infty} \frac{\beta e^{\beta u_i}}{\sum_j e^{\beta u_j}} &= \lim_{\beta \rightarrow \infty} \frac{\beta e^{-\beta \Delta_i}}{\sum_j e^{-\beta \Delta_j}} \\
&= \frac{1}{\Delta_i} \lim_{\beta \rightarrow \infty} \frac{\beta \Delta_i e^{-\beta \Delta_i}}{\sum_{j \in \mathcal{A}} e^{-\beta \Delta_i} + \sum_{j \notin \mathcal{A}} e^{-\beta \Delta_j}} \\
&= \frac{1}{\Delta_i} \cdot \frac{\lim_{\beta \rightarrow \infty} \beta \Delta_i e^{-\beta \Delta_i}}{A + \lim_{\beta \rightarrow \infty} e^{-\beta \Delta_j}} \\
&= 0
\end{aligned}$$

where the first equality comes from multiplying the top and bottom by $e^{-\beta M}$ where $M := \max\{u_1, \dots, u_N\}$, and the final equality again comes from the fact that $\lim_{u \rightarrow \infty} u e^{-u} = 0$. Putting the pieces together, we have that

$$\lim_{\beta \rightarrow \infty} \frac{e^{\beta u_i}}{\sum_j e^{\beta u_j}} \left[1 + \beta(u_i - \mathbf{sm}^\beta(u)) \right] = \left(\lim_{\beta \rightarrow \infty} \frac{\beta e^{\beta u_i}}{\sum_j e^{\beta u_j}} \right) \cdot \left(\lim_{\beta \rightarrow \infty} \{u_i - \mathbf{sm}^\beta(u)\} \right) = 0 \cdot \Delta_i = 0,$$

thus completing the proof. □

In the sequel, we will want to compare $\mathbf{sm}_i^\beta(u_i)$ to $\max_i u_i$ in order to evaluate bias and second order remainders. We start with a bias bound. The following lemma is an analogue of Lemma 6 from Chen et al. [2023], which concerns bounding softmax bias under certain linear feature embeddings. Since we need not worry about such embeddings in our non-parametric setting, we arrive a simple, more streamlined proof.

Lemma A.2. *Let $\beta \geq 0$ be a temperature and let $U_1, \dots, U_N$ be random variables. Define the random differences $\Delta_i := \max_i U_i - \mathbf{sm}_i^\beta U_i \geq 0$. We have the bound*

$$\mathbb{E} \left[\max_i U_i - \mathbf{sm}_i^\beta U_i \right] \leq \sum_{i=1}^N \mathbb{E} [\Delta_i \exp \{-\beta \Delta_i\}].$$

Proof. Let $M := \max_i U_i$ and recall $w_i^\beta(u) := \frac{e^{\beta U_i}}{\sum_{j=1}^N e^{\beta U_j}}$. First, observe that the weight $w_i^\beta(u)$ can be bounded as

$$w_i^\beta(u) = \frac{e^{-\beta M}}{e^{-\beta M}} \frac{e^{\beta U_i}}{\sum_j e^{\beta U_j}} = \frac{e^{-\beta \Delta_i}}{\sum_j e^{-\beta \Delta_j}} \leq e^{-\beta \Delta_i},$$

where the final inequality follows since the sum in the denominator contains $e^{-\beta \Delta_{i^*}} = 1$, where $i^* = \arg \max(u)$. From this, we can deduce result via the following reasoning.

$$M - \mathbf{sm}_i^\beta U_i = \sum_{i=1}^N w_i^\beta(U) \Delta_i \leq \sum_{i=1}^N \Delta_i e^{-\beta \Delta_i}.$$

Taking expectations furnishes the desired result. $\square$

The above lemma allows us to bound the bias by an exponentially downweighted sum of the sub-optimality gaps. The following lemma from Chen et al. [2023] allows us to bound the expectations of these gaps using the rate of growth of the density of each Δ_i in a small neighborhood of zero.

Lemma A.3. *Let U be a non-negative random variable, and suppose there are constants $c, H > 0$ such that $U \mathbb{1}\{U \in (0, c)\}$ admits a Lebesgue density f_U satisfying $f_U(u) \leq H u^{\delta-1}$. Then, there exists a constants $C, \beta_* > 0$ only depending on c, H , and δ such that*

$$\mathbb{E}[U \exp(-\beta U)] \leq C \left(\frac{1}{\beta} \right)^{1+\delta} \quad \forall \beta \geq \beta_*.$$

Proof. While Lemma 7 of Chen et al. [2023] is only stated for $\delta \in (0, 1]$, in the paper the authors show the more general upper bound

$$\mathbb{E}[U \exp(-\beta U)] \leq H \left(\frac{1}{\beta} \right)^{1+\delta} \int_0^\infty v^\delta e^{-v} dv + \frac{(1+\epsilon) \log(\beta)}{\beta^{2+\epsilon}}$$

where $\epsilon > 0$ is arbitrary and $\beta > 0$ is sufficiently large such that $(1+\epsilon) \log(\beta)/\beta < c$ and $(1+\epsilon) \log(\beta) > 1$. Recognizing that the final integral is just $\Gamma(\delta+1)$, where Γ is the Gamma function, we can rewrite the above as

$$\mathbb{E}[U \exp(-\beta U)] \leq H \Gamma(\delta+1) \left(\frac{1}{\beta} \right)^{1+\delta} + \frac{(1+\epsilon) \log(\beta)}{\beta^{2+\epsilon}},$$

Thus, if we simply select $\epsilon_* > \delta - 1$, there exists $\beta_* > 0$ such that the two stated conditions on β hold and such that $\frac{(1+\epsilon_*) \log(\beta)}{\beta^{2+\epsilon_*}} \leq \left(\frac{1}{\beta} \right)^{1+\delta}$. Thus, the claimed bound holds with $C = H \Gamma(1+\delta) + 1$. $\square$

B Proofs from Section 3

The proofs from Section 3 largely follow as corollaries of our more general results from Section 4, namely Proposition 4.1 and Theorem 4.2. Nonetheless, we provide short proofs for each result. We start with arguing the orthogonality of the smoothed score for the value of the optimal treatment policy.

Proof of Proposition 3.2. To prove this result, we just need to check that the conditions of Assumption 2 from Section 2 follow from Assumption 1. Here, the k th score $\psi_k(X; g_k)$ is simply given by $\psi_k(X; g_k) = g_k(k, X)$, where $g_k^*(a, x) := Q^*(a, x) = \mathbb{E}[Y \mid X = x, A = a]$. Checking the first point of the assumption follows immediately from this characterization. Likewise, to check the second point, we see that we have

$$D_Q \psi_\ell(X; g_\ell^*) = D_Q Q^*(\ell, X)(\omega) = \omega(\ell, X) = \mathbb{E} \left[\frac{\mathbb{1}\{A = \ell\}}{p^*(\ell \mid X)} \omega(A, X) \mid X \right].$$

Now we check the third and final point. Observe that for any x and $\nu : \mathcal{W} \rightarrow \mathbb{R}^N$, we that

$$\ell_{x,\nu}(\epsilon) = \psi(x; Q^* + \epsilon\nu) = Q^*(\cdot, x) + \epsilon\nu(\cdot, x)$$

is continuous in ϵ , where $Q^*(\cdot, x) := (Q^*(a, x) : a \in [N])$ and $\nu(\cdot, x) = (\nu(a, x) : a \in [N])$. Observing now that

$$\partial_\epsilon \ell_{x,\nu}(\epsilon) = \nu(\cdot, x)$$

yields continuity of the first derivative. Finally, since ψ is linear in Q , we have $\partial_\epsilon^2 \ell_{x,\nu}(\epsilon) = 0$. Thus, the first part of the third assumption is satisfied. Further, under the assumption that $\|\nu\|_{L^2(P_W)}(P_W) \leq \infty$ and assuming strong positivity, we have for any $C > 0$ that

$$\sup_{|\epsilon| \leq C} \|\partial_\epsilon \ell_{x,\nu}(\epsilon)\|_2 = \|\nu(\cdot, X)\|_2 := F_{\nu,C}.$$

Now, we see that

$$\mathbb{E} F_{\nu,C} \leq \sqrt{\sum_{k=1}^N \mathbb{E} \nu(k, X)^2} \lesssim \sqrt{\mathbb{E} \nu(A, X)^2} < \infty,$$

where the second to last inequality follows from strong positivity. Thus, the second point of the third assumption holds. We can now invoke Proposition 4.1 from Section 4 to yield the desired result. $\square$

Next, we need to check that the assumptions of Theorem 3.3 and Assumption 1 imply those of the more general result presented in Theorem 4.2.

Proof of Theorem 3.3. We iterate through each of the six assumptions of Theorem 4.2. The first and second assumptions of Theorem 4.2 are the same as those of Theorem 3.3. Next, we verify that strong positivity implies mean-squared continuity with $\gamma = 1$. In particular, one checks that we have

$$\begin{aligned} \|\psi_\ell(X; g_\ell) - \psi_\ell(X; g'_\ell)\|_{L^2(P_X)} &= \|Q(\ell, X) - Q'(\ell, X)\|_{L^2(P_X)} \\ &= \mathbb{E} [(Q(\ell, X) - Q'(\ell, X))^2]^{1/2} \\ &= \mathbb{E} \left[\frac{\mathbb{1}\{A = \ell\}}{p^*(\ell \mid X)} (Q(\ell, X) - Q'(\ell, X))^2 \right]^{1/2} \\ &\leq \frac{1}{\eta} \mathbb{E} [(Q(A, X) - Q'(A, X))^2]^{1/2} \\ &\lesssim \|Q - Q'\|_{L^2(P_W)}. \end{aligned}$$

We note that the assumptions on the nuisance convergence rates are identical (the fourth condition) and likewise noting that $U_k = Y$, the assumptions on boundedness (the fifth condition) are identical.

We conclude by verifying the sixth assumption. First, we check the condition placed on the expected square of the first Gateaux derivative. In particular, for any $k \in [N]$, we have

$$\begin{aligned} \mathbb{E}_X \left[\left(D_{Q_k} \psi_k(X; \bar{Q})(\hat{Q}_n - Q^*) \right)^2 \right] &= \mathbb{E}_X \left[(\hat{Q}_n(k, X) - Q^*(k, X))^2 \right] \\ &\lesssim \mathbb{E}_Z [(\hat{Q}_n(A, X) - Q^*(A, X))^2] \quad (\text{Strong Positivity}). \end{aligned}$$

Likewise, since $\psi_k(x; Q)$ is linear in Q , the expected second Gateaux derivative is identically zero. This completes the proof, as we can now invoke Theorem 4.2 to give the desired asymptotic normality result. $\square$

Remark B.1. In the case of two actions $A = 1$ (treatment) and $A = 0$ (control), one can arrive at the at first simpler seeming Neyman orthogonal score

$$\Phi^\beta(z; Q, \alpha) = \varphi_\beta(\tau(X)) + Q(0, x) - \alpha(a, x)(y - Q(a, x)),$$

where $\tau(x) = Q(1, x) - Q(0, x)$ and $\varphi_\beta(u) := u\sigma(\beta u)$. Here, the additional nuisance/Riesz representer is

$$\alpha^\beta(a, x) = \left(\frac{a}{p^*(x)} - \frac{1-a}{1-p^*(x)} \right) \varphi'_\beta(\tau^*(x)) + \frac{1-a}{1-\pi_0(x)},$$

where $p^*(x) := \mathbb{P}(A = 1 \mid X = x)$. However, through noting the identities

$$\mathbf{sm}^\beta(a, b) = (b - a)\sigma(\beta(b - a)) + a$$

and

$$\partial_a \mathbf{sm}^\beta(a, b) = 1 - \varphi'_\beta(b - a) \quad \text{and} \quad \partial_b \mathbf{sm}^\beta(a, b) = \varphi'_\beta(b - a),$$

we actually have the equivalence $\alpha^\beta(a, x) = \alpha_0^\beta(a, x) + \alpha_1^\beta(a, x)$ where α_0^β and α_1^β are as defined in Proposition 3.2. Further, we actually have

$$\Phi^\beta(z; Q, \alpha) = \Psi^\beta(z; Q, \alpha_0, \alpha_1),$$

where $\alpha_0(a, x) := (1 - a)\alpha(a, x)$ and $\alpha_1(a, x) = a\alpha(a, x)$, so the scores are actually equivalent. We choose the work with the score Ψ^β as it is more general, allowing for an arbitrary finite number of actions.

C Proofs from Section 4

We start by proving Proposition 4.1, which provides a general de-biased smoothed score for the parameters specified as an appropriate maximum of scores.

Proof of Proposition 4.1. First, we note that the fact that $V^\beta = \mathbb{E}[\Psi^\beta(Z; g^*, \alpha^\beta)]$ is immediate, as for any $k \in [N]$ we have via the tower rule for conditional expectations that

$$\mathbb{E}[\alpha_k^\beta(W)^\top (U_k - g_k^*(W))] = \mathbb{E}[\alpha_k^\beta(W)^\top (\mathbb{E}[U_k \mid W] - g_k^*(W))] = 0.$$

Next, we check Neyman orthogonality. Let $\omega \in L^2(P_W)$ be arbitrary. By Assumption 2, for any $C > 0$, there is an integrable random variable $F_{\omega, C}$ such that $\sup_{|\epsilon| \leq C} |\partial_\epsilon \ell_{\omega, X}(\epsilon)| \leq F_{\omega, C}$ almost surely. Fixing $C = 1$ and using the same analysis as in the proof of Lemma C.3 below, we see that

1. $D_{g_k} \mathbb{E}[\mathbf{sm}^\beta \psi(X; g^*)](\omega) = \lim_{\substack{t \rightarrow 0 \\ |t| \leq 1}} \mathbb{E}[D_{g_k} \mathbf{sm}^\beta \psi(X; \tilde{g}_t)(\omega)]$ where $\tilde{g}_t \in [g^*, g^* + t\omega]$ for each t ,
2. $\mathbf{sm}^\beta \psi(X; \tilde{g}_t) \xrightarrow[t \rightarrow 0]{a.s.} \mathbf{sm}^\beta \psi(X; g^*)$ (since $g^* + t\omega \rightarrow g^*$ a.s.), and
3. $|D_{g_k} \mathbf{sm}^\beta \psi(X; \tilde{g}_t)(\omega)| \lesssim \sup_{|\epsilon| \leq 1} |\partial_\epsilon \ell_{X, \omega}(\epsilon)| \leq F_{\omega, 1}$

Consequently, the dominated convergence theorem yields that $D_{g_k} \mathbb{E}[\mathbf{sm}^\beta \psi(X; g^*)](\omega) = \mathbb{E}[D_{g_k} \mathbf{sm}^\beta \psi(X; g^*)(\omega)]$, and thus we can switch the order of differentiation and expectation. Following from this, we have that

$$\begin{aligned} D_{g_k} \mathbb{E}[\mathbf{sm}_\ell^\beta \psi(X; g_\ell^*)](\omega) &= \mathbb{E}[D_{g_k} \mathbf{sm}_\ell^\beta \psi_\ell(X; g_\ell^*)(\omega)] && \text{(Switching limits via Lemma C.3)} \\ &= \mathbb{E}[\partial_k \mathbf{sm}_\ell^\beta \psi_\ell(X; g_\ell^*) D_{g_k} \psi_k(X; g_k^*)(\omega)] && \text{(Chain rule)} \\ &= \mathbb{E}[\partial_k \mathbf{sm}_\ell^\beta \psi_\ell(X; g_\ell^*) \mathbb{E}(\zeta_k^*(W)^\top \omega(W) \mid X)] && \text{(Assumption 2 Part 2)} \\ &= \mathbb{E}[\partial_k \mathbf{sm}_\ell^\beta \psi_\ell(X; g_\ell^*) \zeta_k^*(W)^\top \omega(W)] && \text{(Tower rule)} \\ &= \mathbb{E}[\alpha_k^\beta(W)^\top \omega(W)] && \text{(Definition of } \alpha_k^\beta). \end{aligned}$$

Now, we see that

$$\begin{aligned}
D_{g_k} \mathbb{E}[\alpha_k^\beta(W)^\top (U_k - g_k^*(W))](\omega) &= -D_{g_k} \mathbb{E} \left[\alpha_k^\beta(W)^\top g_k^*(W) \right] (\omega) \\
&= -\partial_t \mathbb{E} \left[\alpha_k^\beta(W)^\top (g_k^*(W) + t\omega(W)) \right] \\
&= -\lim_{t \rightarrow 0} \mathbb{E} \left[\frac{\alpha_k^\beta(W)^\top (g_k^*(W) + t\omega(W)) - \alpha_k^\beta(W)^\top g_k^*(W)}{t} \right] \\
&= -\mathbb{E}[\alpha_k^\beta(W)^\top \omega(W)]
\end{aligned}$$

and so the Gateaux derivative of $\mathbb{E}[\Psi^\beta(Z; g^*, \alpha^\beta)]$ with respect to each g_k vanishes. Checking the Gateaux derivative with respect to α_k is similar, as we have

$$\begin{aligned}
D_{\alpha_k} \mathbb{E}[\Psi^\beta(Z; g^*, \alpha^\beta)](\omega) &= D_{\alpha_k} \mathbb{E}[\alpha_k^\beta(W)^\top (U_k - g_k^*(W))](\omega) \\
&= \mathbb{E}[\omega(W)^\top (U_k - g_k^*(W))] \\
&= \mathbb{E}[\omega(W)^\top (\mathbb{E}(U_k | W) - g_k^*(W))] \\
&= 0.
\end{aligned}$$

Thus, we have shown that the score is Neyman orthogonal. $\square$

Next, we check our main asymptotic linearity/normality result for smoothed estimates of irregular functionals. This boils down to checking the conditions of the smoothed central limit theorem presented in Theorem D.1 of Appendix D.

Proof. First, we show we have

$$\sqrt{n}(\widehat{V}_n - V^*) = \sqrt{n}(\widehat{V}_n - V^{\beta_n}) + \underbrace{\sqrt{n}(V^{\beta_n} - V^*)}_{\text{Bias}(\beta_n)} = \sqrt{n}(\widehat{V}_n - V^{\beta_n}) + o_{\mathbb{P}}(1).$$

To show this, it is sufficient to show that $\text{Bias}(\beta_n) = o(n^{-1/2})$. Under the specific choice of sequence of $(\beta_n)_{n \geq 1}$, applying Lemmas A.3 and A.2 yields, for β sufficiently large,

$$\begin{aligned}
\text{Bias}(\beta_n) &= \left| \mathbb{E}[\mathbf{sm}_\ell^\beta(\psi_\ell(X; g_\ell^*)) - \max_\ell \{\psi_\ell(X; g_\ell^*)\}] \right| \\
&\leq \sum_{k=1}^N \mathbb{E}[\Delta_k \exp\{-\beta_n \Delta_k\}] \\
&\lesssim \left(\frac{1}{\beta_n} \right)^{1+\delta} \\
&= o(n^{-1/2}) \quad \left(\text{Since } \beta_n = \omega \left(n^{-\frac{1}{2(1+\delta)}} \right) \right).
\end{aligned}$$

Thus, we just need to show asymptotic linearity off $\sqrt{n}(\widehat{V}_n - V^{\beta_n})$ to prove the result. It suffices to check the conditions of Theorem D.1.

Checking Condition 1 The first condition is satisfied as $\Psi^\beta(Z; g, \alpha)$ is Neyman orthogonal by construction (see Proposition 4.1).

Checking Condition 2 Letting $h := (g, \alpha)$ and letting $\bar{g} \in [\widehat{g}_n, g^*]$ and $\bar{\alpha} \in [\widehat{\alpha}_n, \alpha^{\beta_n}]$, one can compute that

$$D_h^2 \mathbb{E}_Z \Psi^{\beta_n}(Z; \bar{g}, \bar{\alpha})(\widehat{h} - h^{\beta_n}) = D_{\bar{g}}^2 \mathbb{E}_Z \left[\mathbf{sm}_\ell^{\beta_n}(\psi_\ell(X; \bar{g}_\ell)) \right] (\widehat{g}_n - g^*) + 2D_{g, \alpha} \mathbb{E}_Z \left[\sum_{k=1}^N \bar{\alpha}_k(W)^\top (U_k - \bar{g}_k(W)) \right] (\widehat{h}_n - h^{\beta_n}),$$

so it suffices to show that each of the terms on the right hand side are of order $o_{\mathbb{P}}(n^{-1/2})$. In the sequel, we leverage the following chain rule for second Gateaux derivatives²: given a twice differentiable function $F : \mathbb{R}^N \rightarrow \mathbb{R}$ and a twice Gateaux differentiable function $H : L^2(P_W) \rightarrow \mathbb{R}^N$, one has,

$$D_g^2(F \circ H)(g)(\omega) = D_g H(g)(\omega)^\top \nabla_u^2 F(H(g)) D_g H(g)(\omega) + \nabla_u F(H(g))^\top D_g^2 H(g)(\omega), \quad (4)$$

which follows from the same argument used to prove the multivariate chain rule (see Skorski [2019]).

Analyzing the first term, we see we have

$$\begin{aligned} D_g^2 \mathbb{E}_Z \left[\mathbf{sm}_\ell^{\beta_n}(\psi_\ell(X; \bar{g}_\ell)) \right] (\hat{g}_n - g^*) &= \mathbb{E}_Z \left[D_g^2 \mathbf{sm}_\ell^{\beta_n} \psi_\ell(X; \bar{g}_\ell) (\hat{g}_n - g^*) \right] && \text{(By Assumption 6)} \\ &= \mathbb{E}_Z \left[\underbrace{D_g \psi(X; \bar{g}) (\hat{g}_n - g^*)^\top \nabla^2 \mathbf{sm}_\ell^{\beta_n} \psi_\ell(X; \bar{g}_\ell) D_g \psi(X; \bar{g}) (\hat{g}_n - g^*)}_{T_1} \right] \\ &\quad + \mathbb{E}_Z \left[\underbrace{\nabla_u \mathbf{sm}_\ell^{\beta_n} \psi_\ell(X; \bar{g}_\ell)^\top D_g^2 \psi(X; \bar{g}) (\hat{g}_n - g^*)}_{T_2} \right] && \text{(Using Equation (4)).} \end{aligned}$$

We now show that each of the two terms above grows as $o_{\mathbb{P}}(n^{-1/2})$ under the nuisance estimation rates assumed in the theorem. First, we can control T_1 as follows:

$$\begin{aligned} T_1 &\leq \sup_{u \in \mathbb{R}^N} \|\nabla_u^2 \mathbf{sm}_\ell^{\beta_n}(u)\|_{op} \cdot \mathbb{E}_Z \|D_g \psi(X; \bar{g}) (\hat{g}_n - g^*)\|_2^2 && \text{(Since } |x^\top A x| \leq \|A\|_{op} \|x\|_2^2) \\ &= \sup_{u \in \mathbb{R}^N} \|\nabla_u^2 \mathbf{sm}_\ell^{\beta_n}(u)\|_{op} \sum_{k=1}^N \|D_{g_k} \psi_k(X; \bar{g}_k)\|_{L^2(P_X)}^2 \\ &\lesssim \sup_{u \in \mathbb{R}^N} \|\nabla_u^2 \mathbf{sm}_\ell^{\beta_n}(u)\|_{op} \sum_{k=1}^N \|\hat{g}_{n,k} - g_k^*\|_{L^2(P_W)}^2 && \text{(Assumption 6)} \\ &\lesssim \beta_n \sum_{k=1}^N \|\hat{g}_{n,k} - g_k^*\|_{L^2(P_W)}^2 && \text{(By Lemma A.1 Point 3)} \\ &= o_{\mathbb{P}}(n^{-1/2}) && \text{(Since } \|\hat{g}_{n,k} - g_k^*\|_{L^2(P_W)} = o_{\mathbb{P}}(\beta_n n^{-1/2}) \text{).} \end{aligned}$$

Next, we bound T_2 . We have

$$\begin{aligned} T_2 &= \mathbb{E}_Z \left[\nabla_u \mathbf{sm}_\ell^{\beta_n} \psi_\ell(X; \bar{g}_\ell)^\top D_g^2 \psi(X; \bar{g}) (\hat{g}_n - g^*) \right] \\ &= \sum_{k=1}^N \mathbb{E}_Z \left[\partial_k \mathbf{sm}_\ell^{\beta_n} \psi(X; \bar{g}) \cdot D_{g_k}^2 \psi_k(X; \bar{g}_k) (\hat{g}_{n,k} - g_k^*) \right] \\ &\leq \sum_{k=1}^N \|\nabla_u \mathbf{sm}_\ell^{\beta_n} \psi(X; \bar{g})\|_{L^\infty(P_X)} \cdot \mathbb{E}_X \left[D_{g_k}^2 \psi_k(X; \bar{g}_k) (\hat{g}_{n,k} - g_k^*) \right] && (L^1/L^\infty \text{ Holder's inequality}) \\ &\lesssim \sum_{k=1}^N \mathbb{E} \left[D_{g_k}^2 \psi_l(X; \bar{g}_k) (\hat{g}_{n,k} - g_k^*) \right] && \text{(Lemma A.1 Part 1)} \\ &\lesssim \sum_{k=1}^N \|\hat{g}_n - g^*\|_{L^2(P_W)}^2 && \text{(Assumption 6)} \\ &= o_{\mathbb{P}}(n^{-1/2}) && (\|\hat{g}_{n,k} - g_k^*\|_{L^2(P_W)}^2 = o_{\mathbb{P}}(n^{-1/2}) \quad \forall k). \end{aligned}$$

Likewise, we can check the cross-derivative term. Let $\Delta_{g,k} := \hat{g}_{n,k} - g_k^*$ and $\Delta_{\alpha,k} := \hat{\alpha}_{n,k} - \alpha_k^{\beta_n}$. Analyzing

²Here and throughout, by $\nabla_u^2 F(H(g))$ we actually mean $\nabla_u F(u)|_{u=H(g)}$.

each term in the finite sum yields:

$$\begin{aligned}
& D_{g_k, \alpha_k} \mathbb{E}_Z [\bar{\alpha}_k(W)^\top (U_k - \bar{g}_k(W))] (\Delta_{g,k}, \Delta_{\alpha,k}) \\
&= -\frac{\partial^2}{\partial s \partial t} \mathbb{E}_Z [(\bar{\alpha}_k + t \Delta_{\alpha,k})^\top (\bar{g}_k + s \Delta_{g,k})] \Big|_{s,t=0} \\
&= -\lim_{s,t \rightarrow 0} \mathbb{E}_Z \left[\frac{(\bar{\alpha}_k + t \Delta_{\alpha,k})^\top (\bar{g}_k + s \Delta_{g,k}) - \bar{\alpha}_k^\top (\bar{g}_k + s \Delta_{g,k}) - (\bar{\alpha}_k + t \Delta_{\alpha,k})^\top \bar{g}_k + \bar{\alpha}_k^\top \bar{g}_k}{st} \right] \\
&= -\lim_{s,t \rightarrow 0} \mathbb{E}_Z \left[\frac{st \Delta_{\alpha,k} \Delta_{g,k}}{st} \right] = -\mathbb{E}_Z [\Delta_{\alpha,k} \Delta_{g,k}],
\end{aligned}$$

where outside of the first line above we have suppressed dependence on W for notational ease. Leveraging this identity, we have that:

$$\begin{aligned}
|D_{g_k, \alpha_k} \mathbb{E}_Z [\bar{\alpha}_k(W)^\top (U_k - \bar{g}_k(W))] (\Delta_{g,k}, \Delta_{\alpha,k})| &= \left| \langle \hat{\alpha}_{k,n} - \alpha_k^{\beta_n}, \hat{g}_{k,n} - g_k^* \rangle_{L^2(P_W)} \right| \\
&\leq \|\hat{\alpha}_{k,n} - \alpha_k^{\beta_n}\|_{L^2(P_W)} \cdot \|\hat{g}_{k,n} - g_k^*\|_{L^2(P_W)} \\
&= o_{\mathbb{P}}(n^{-1/2}).
\end{aligned}$$

Thus, we have show that, under the assumptions of Theorem 4.2, second order contributions vanish at sufficiently fast rates.

Checking Condition 3 We show the stronger result that $\Psi^{\beta_n}(z; g^*, \alpha^{\beta_n}) \xrightarrow{n \rightarrow \infty} \Psi^*(z; g^*, \alpha^*)$ for all z , which in turns proves convergence in probability. First, applying Point 4 of Lemma A.1, we see that

$$\lim_{n \rightarrow \infty} \mathbf{sm}_\ell^{\beta_n} \psi_\ell(x; g_\ell^*) = \lim_{\beta \rightarrow \infty} \mathbf{sm}_\ell^\beta \psi_\ell(x; g_\ell^*) = \max_\ell (\psi_\ell(x; g_\ell^*)).$$

Next, Point 5 of the same lemma implies that, for any $k \in [N]$,

$$\lim_{n \rightarrow \infty} \alpha_k^{\beta_n}(w)^\top (u_k - g_k^*(w)) = \left(\lim_{\beta \rightarrow \infty} \partial_k \mathbf{sm}_\ell^\beta \psi(z; g_\ell^*) \right) \zeta_k^*(w)^\top (u_k - g_k^*(w)) = \mathfrak{s}_k(w) \zeta_k^*(w)^\top (u_k - g_k^*(w)).$$

The result readily follows from this.

Checking Condition 4 We now check stochastic equi-continuity. In particular, we aim to show:

$$|\mathbb{G}_n \Psi^{\beta_n}(Z; \hat{g}_n, \hat{\alpha}_n) - \mathbb{G}_n \Psi^{\beta_n}(Z; g^*, \alpha^{\beta_n})| = o_{\mathbb{P}}(1).$$

To do this, we start by showing:

$$\|\mathbb{G}_n \Psi^{\beta_n}(Z; g, \alpha) - \mathbb{G}_n \Psi^{\beta_n}(Z; g', \alpha')\|_{L^2(P_Z)} \lesssim \sum_\ell \|g_\ell - g'_\ell\|_{L^2(P_W)} + \sum_\ell \|\alpha_\ell - \alpha'_\ell\|_{L^2(P_W)} + \sum_\ell \|g_\ell - g'_\ell\|_{L^2(P_W)}^\gamma$$

for any g, g', α, α' that have their $L^\infty(P_W)$ norm bounded by the constant B in the theorem statement.

For ease of notation, define $f(Z; h, h') := \Psi^{\beta_n}(Z; h) - \Psi^{\beta_n}(Z; h')$ where we have let $h = (g, \alpha)$ and $h' = (g', \alpha')$. First, observe that we have

$$\begin{aligned}
\|\mathbb{G}_n \Psi^{\beta_n}(Z; h) - \mathbb{G}_n \Psi^{\beta_n}(Z; h')\|_{L^2(P_Z)} &= \frac{1}{\sqrt{n}} \left(\mathbb{E} \left(\sum_{i=1}^n \{f(Z_i; h, h') - \mathbb{E} f(Z; h, h')\} \right)^2 \right)^{1/2} \\
&= \frac{1}{\sqrt{n}} \left(\sum_{i=1}^n \text{Var}[f(Z; h, h')] \right)^{1/2} && \text{(Independence of the } Z_i) \\
&\leq \frac{1}{\sqrt{n}} \left(\sum_{i=1}^n \mathbb{E} f(Z; h, h')^2 \right)^{1/2} && \text{(Since } \text{Var}[X] \leq \mathbb{E}[X^2]) \\
&= \mathbb{E}[f(Z; g, g')^2]^{1/2} = \|\Psi^{\beta_n}(Z; h) - \Psi^{\beta_n}(Z; h')\|_{L^2(P_Z)}.
\end{aligned}$$

We have:

$$\begin{aligned} \|\Psi^{\beta_n}(Z; h') - \Psi^{\beta_n}(Z; h)\|_{L^2(P_Z)} &\leq \|\mathbf{sm}_\ell^\beta \psi_\ell(X; g_\ell) - \mathbf{sm}_\ell^\beta \psi_\ell(X; g'_\ell)\|_{L^2(P_X)} \\ &\quad + \sum_\ell \|\alpha_\ell(W)^\top (U_\ell - g_\ell(W)) - \alpha'_\ell(W)^\top (U_\ell - g'_\ell(W))\|_{L^2(P_x)}. \end{aligned}$$

Letting $\psi(X; g) := (\psi_1(X; g_1), \dots, \psi_N(X; g_N))$, the first term can be bounded above by

$$\begin{aligned} \|\mathbf{sm}_\ell^\beta \psi_\ell(X; g_\ell) - \mathbf{sm}_\ell^\beta \psi_\ell(X; g'_\ell)\|_{L^2(P_X)} &\leq L \|\psi(X; g) - \psi(X; g')\|_{L^2(P_X)} \\ &\leq L \sum_\ell \|\psi_\ell(X; g_\ell) - \psi_\ell(X; g'_\ell)\|_{L^2(P_X)} \\ &\lesssim \sum_\ell \|g_\ell - g'_\ell\|_{L^2(P_W)}^\gamma \quad (\text{Mean-squared continuity}). \end{aligned}$$

where L is the β -independent Lipschitz constant for $\mathbf{sm}^\beta$ outlined in Lemma A.1. An arbitrary index in the second term can likewise be bounded as:

$$\begin{aligned} &\|\alpha_\ell(W)^\top (U_\ell - g_\ell(W)) - \alpha'_\ell(W)^\top (U_\ell - g'_\ell(W))\|_{L^2(P_Z)} \\ &= \|\alpha_\ell(W)^\top (U_\ell - g_\ell(W)) \pm \alpha_\ell(W)^\top (U_\ell - g'_\ell(W)) - \alpha'_\ell(W)^\top (U_\ell - g'_\ell(W))\|_{L^2(P_Z)} \\ &\leq \|\alpha_\ell(W)^\top (g_\ell - g'_\ell(W))\|_{L^2(P_W)} + \|(\alpha_\ell - \alpha'_\ell)(W)^\top (U_\ell - g'_\ell(W))\|_{L^2(P_Z)} \\ &\lesssim \|\alpha_\ell - \alpha'_\ell\|_{L^2(P_W)} + \|g_\ell - g'_\ell\|_{L^2(P_W)}, \end{aligned}$$

where the second to last inequality follows from the triangle inequality at the final inequality follows from the boundedness assumption on α, α', g, g' , and U_ℓ . This in total yields:

$$\|\Psi^{\beta_n}(X; g') - \Psi^{\beta_n}(X; g)\|_{L^2(P_X)} \lesssim \underbrace{\sum_\ell \|g_\ell - g'_\ell\|_{L^2(P_W)} + \sum_\ell \|\alpha_\ell - \alpha'_\ell\|_{L^2(P_W)} + \sum_\ell \|g_\ell - g'_\ell\|_{L^2(P_W)}^\gamma}_{\Delta(h, h')}. \quad (5)$$

Now, note that, by the assumption of bounded true nuisances and nuisance estimates (Assumption 5 of the theorem) that $|\Delta(\hat{h}_n, h^{\beta_n})^2| \leq B'$ for any $n \geq 1$ where $B' > 0$ is some absolute constant. Consequently, the process $(\Delta_n(\hat{h}_n, h^{\beta_n}))_{n \geq 1}$ is uniformly integrable. Since, by the assumption of nuisance consistency, we have $\Delta(\hat{h}_n, h^{\beta_n}) = o_{\mathbb{P}}(1)$, we have that $\lim_{n \rightarrow \infty} \mathbb{E}[\Delta(\hat{h}_n, h^{\beta_n})] = 0$.³ From this, we have by applying Chebyshev's inequality conditionally on $(\hat{g}_n, \hat{\alpha}_n)$ that for any $\delta \in (0, 1)$:

$$\begin{aligned} &\lim_{n \rightarrow \infty} \mathbb{P}\left(|\mathbb{G}_n \Psi^{\beta_n}(Z; \hat{h}_n) - \mathbb{G}_n \Psi^{\beta_n}(Z; h^{\beta_n})| \geq \delta\right) \\ &\leq \lim_{n \rightarrow \infty} \frac{1}{\delta^2} \mathbb{E}\left[\left(\sum_\ell \|g_\ell - g'_\ell\|_{L^2(P_W)} + \sum_\ell \|\alpha_\ell - \alpha'_\ell\|_{L^2(P_W)} + \sum_\ell \|g_\ell - g'_\ell\|_{L^2(P_W)}^\gamma\right)^2\right] \\ &= \lim_{n \rightarrow \infty} \frac{1}{\delta^2} \mathbb{E}\left[\Delta(\hat{h}_n, h^{\beta_n})^2\right] = 0. \end{aligned}$$

This proves the desired stochastic equi-continuity result.

Checking Condition 5 Regularity conditions are straightforward to check. First, observe that we have

$$|V^\beta| \leq \mathbb{E}|\mathbf{sm}_\ell^\beta \psi_\ell(X; g_\ell^*)| \leq \mathbb{E}|\max_\ell \psi_\ell(X; g_\ell^*)| \leq B$$

by assumption. Next, similar reasoning yields that, for any $\epsilon > 0$,

$$\|\Psi^{\beta_n}(X; g^*, \alpha^{\beta_n})\|_{L^{2+\epsilon}(P_Z)} \leq \|\Psi^{\beta_n}(Z; g^*, \alpha^{\beta_n})\|_{L^\infty(P_Z)} \leq B + 2B^2.$$

This completes the proof. $\square$

³This follows from the fact that, if $(X_n)_{n \geq 1}$ is uniformly integrable and $X_n \rightarrow X$, then $\lim_{n \rightarrow \infty} |\mathbb{E}[X_n] - \mathbb{E}[X]| = 0$.

We now prove Corollary 4.3 by showing an appropriate level of consistency for the plug-in estimate $\widehat{\Sigma}_n$ of the population variance Σ^* . In the following proof, we need the following weak law of large numbers for triangular arrays. The following appears as Theorem 2.2.6 in [Durrett, 2019], wherein a proof can be found.

Proposition C.1. *For each $n \geq 1$, let $X_1^n, \dots, X_n^n$ be a sequence of random variables, $S_n := X_1^n + \dots + X_n^n$, $\sigma_n^2 := \text{Var}[S_n]$, $\mu_n := \mathbb{E}S_n$, and b_n a sequence of constants. Then, if $\sigma_n^2/b_n^2 \rightarrow 0$ as $n \rightarrow \infty$, we have*

$$\frac{S_n - \mu_n}{b_n} \xrightarrow{\mathbb{P}} 0.$$

Proof of Corollary 4.3. Observe that we can equivalently write

$$\widehat{\Sigma}_n := \frac{1}{n} \sum_m \left\{ \Psi^{\beta_n}(Z; \widehat{h}_n) - \widehat{V}_n \right\}^2 = \frac{1}{n} \sum_m \Psi^{\beta_n}(Z; \widehat{h}_n)^2 - \widehat{V}_n^2.$$

The conclusion of Theorem 4.2 immediately implies that $\widehat{V}_n^2 = (V^*)^2 + o_{\mathbb{P}}(1)$ by the continuous mapping theorem, so it suffices to check that

$$\frac{1}{n} \sum_m \Psi^{\beta_n}(Z; \widehat{h}_n)^2 - \mathbb{E}[\Psi^*(Z; h^*)^2] = o_{\mathbb{P}}(1).$$

We rewrite the left-hand side of the above expression as

$$\begin{aligned} \frac{1}{n} \sum_m \Psi^{\beta_n}(Z; \widehat{h}_n)^2 - \mathbb{E}[\Psi^*(Z; h^*)^2] &= \underbrace{\frac{1}{n} \sum_m \left\{ \Psi^{\beta_n}(Z; \widehat{h}_n)^2 - \Psi^{\beta_n}(Z; h^{\beta_n})^2 \right\}}_{T_1} + \underbrace{\frac{1}{n} \sum_m \left\{ \Psi^{\beta_n}(Z; h^{\beta_n})^2 - \Psi^*(Z; h^*)^2 \right\}}_{T_2} \\ &\quad + \underbrace{\frac{1}{n} \sum_m \Psi^*(Z; h^*)^2 - \mathbb{E}[\Psi^*(Z; h^*)^2]}_{T_3}. \end{aligned}$$

We start by analyzing T_2 and T_3 , as they are both simpler than T_1 . First, since T_3 is an average of i.i.d. bounded mean-zero random variables $\Psi^*(Z; h^*)^2 - \mathbb{E}[\Psi^*(Z; h^*)^2]$, the weak law of large numbers implies that $T_3 = o_{\mathbb{P}}(1)$.

Next, we use a weak law of large numbers for triangular arrays to argue that $T_2 = o_{\mathbb{P}}(1)$. Observe that, by the assumption of boundedness, $\sup_{n \geq 1} \text{Var}[\Psi^{\beta_n}(Z; h^{\beta_n})^2 - \Psi^*(Z; h^*)^2] \leq C$ for some constant C , and hence $\text{Var}[T_2] \leq C/n$. Applying the weak law of large numbers for triangular arrays with $S_n = \sum_m \left\{ \Psi^{\beta_n}(Z; h^{\beta_n})^2 - \Psi^*(Z; h^*)^2 \right\}$ and $b_n = n$ (Proposition C.1) yields that

$$T_2 - \mathbb{E}[\Psi^{\beta_n}(Z; h^{\beta_n})^2 - \Psi^*(Z; h^*)^2] = o_{\mathbb{P}}(1),$$

so it suffices to show that the above expectation goes towards zero as n grows large. In particular, the bounded convergence theorem alongside the fact that $\Psi^{\beta_n}(Z; h^{\beta_n}) \xrightarrow[n \rightarrow \infty]{a.s.} \Psi^*(Z; h^*)$ yields that

$$\lim_{n \rightarrow \infty} \mathbb{E}[\Psi^{\beta_n}(Z; h^{\beta_n})^2 - \Psi^*(Z; h^*)^2] = \mathbb{E}\left[\lim_{n \rightarrow \infty} \Psi^{\beta_n}(Z; h^{\beta_n})^2 - \Psi^*(Z; h^*)^2\right] = 0,$$

which thus yields that $T_2 = o_{\mathbb{P}}(1)$.

Now, we argue that $T_1 = o_{\mathbb{P}}(1)$. Observe that we can rewrite T_1 as follows.

$$\begin{aligned} T_1 &= \frac{1}{n} \sum_m \left\{ \Psi^{\beta_n}(Z; \widehat{h}_n) \pm \Psi^{\beta_n}(Z; h^{\beta_n}) \right\}^2 - \frac{1}{n} \sum_m \Psi^{\beta_n}(Z; h^{\beta_n})^2 \\ &= \frac{1}{n} \sum_m \left\{ \Psi^{\beta_n}(Z; \widehat{h}_n) - \Psi^{\beta_n}(Z; h^{\beta_n}) \right\}^2 + \frac{2}{n} \sum_m \left\{ \Psi^{\beta_n}(Z; \widehat{h}_n) - \Psi^{\beta_n}(Z; h^{\beta_n}) \right\} \cdot \Psi^{\beta_n}(Z; h^{\beta_n}). \end{aligned}$$

Thus, applying the Cauchy-Schwarz inequality, we see that we have

$$|T_1| \leq \frac{1}{n} \sum_m \left\{ \Psi^{\beta_n}(Z; \widehat{h}_n) - \Psi^{\beta_n}(Z; h^{\beta_n}) \right\}^2 + 2 \left(\frac{1}{n} \sum_m \left\{ \Psi^{\beta_n}(Z; \widehat{h}_n) - \Psi^{\beta_n}(Z; h^{\beta_n}) \right\}^2 \right)^{1/2} \left(\frac{1}{n} \sum_m \Psi^{\beta_n}(Z; h^{\beta_n})^2 \right)^{1/2}.$$

Since $\frac{1}{n} \sum_m \Psi^{\beta_n}(Z; h^{\beta_n})^2 = O_{\mathbb{P}}(1)$ by our analysis of T_2 , it suffices to show that $n^{-1} \sum_m \left\{ \Psi^{\beta_n}(Z_m; \hat{h}_n) - \Psi^{\beta_n}(Z_m; h^{\beta_n}) \right\}^2 = o_{\mathbb{P}}(1)$ to show $T_1 = o_{\mathbb{P}}(1)$. First, we note that, in the proof of Theorem 4.2 above, we showed that

$$\mathbb{E}_Z \left[\left(\Psi^{\beta_n}(Z; \hat{h}_n) - \Psi^{\beta_n}(Z; h^{\beta_n}) \right)^2 \right] \lesssim \Delta(\hat{h}_n, h^{\beta_n})^2 = o_{\mathbb{P}}(1),$$

where $\Delta(h, h')$ is as defined in Equation (5). We further argued using uniform integrability that $\lim_{n \rightarrow \infty} \mathbb{E} \left[\Delta(\hat{h}_n, h^{\beta_n})^2 \right] = 0$. Consequently, applying Markov's inequality, we get, for any $\delta > 0$,

$$\begin{aligned} \lim_{n \rightarrow \infty} \mathbb{P} \left(\left| \frac{1}{n} \sum_m \left\{ \Psi^{\beta_n}(Z; \hat{h}_n) - \Psi^{\beta_n}(Z; h^{\beta_n}) \right\}^2 \right| \geq \delta \right) &\leq \frac{1}{\delta} \lim_{n \rightarrow \infty} \mathbb{E} \left[\mathbb{E}_Z \left((\Psi^{\beta_n}(Z; \hat{h}_n) - \Psi^{\beta_n}(Z; h^{\beta_n}))^2 \right) \right] \\ &\lesssim \frac{1}{\delta} \lim_{n \rightarrow \infty} \mathbb{E} \left[\Delta(\hat{h}_n, h^{\beta_n})^2 \right] = 0. \end{aligned}$$

This thus shows that $T_1 = o_{\mathbb{P}}(1)$, and hence $\hat{\Sigma}_n \xrightarrow[n \rightarrow \infty]{\mathbb{P}} \Sigma^*$.

We now show how this suffices to conclude the proof. Observe first that the asymptotic linearity result of Theorem 4.2 can equivalently be written as

$$\sqrt{n}(\Sigma^*)^{-1/2} (\hat{V}_n - V^*) = \frac{1}{\sqrt{n}} \sum_m (\Sigma^*)^{-1/2} \{ \Psi^*(Z_m; h^*) - V^* \} + o_{\mathbb{P}}(1),$$

and the normal sum on the right-hand side converges in distribution to a standard normal random variable. To complete the proof, it suffices to show that

$$\sqrt{n}(\hat{\Sigma}^{-1/2} - (\Sigma^*)^{-1/2})(\hat{V}_n - V^*) = o_{\mathbb{P}}(1).$$

Observe that $\sqrt{n}(\hat{V}_n - V^*) = O_{\mathbb{P}}(1)$ by the conclusion of Theorem 4.2, and since $\Sigma^* > 0$, the continuous mapping theorem alongside the consistency of $\hat{\Sigma}_n$ imply $\hat{\Sigma}_n^{-1/2} - (\Sigma^*)^{-1/2} = o_{\mathbb{P}}(1)$, thus proving the desired result. $\square$

We briefly state a cross-fitting variant of Theorem 4.2. The proof of the following corollary is straightforward, but we nonetheless prove it for completeness.

Corollary C.2. *Suppose the assumptions of Theorem 4.2 hold, and in particular that assumptions 4 and 5 hold for all nuisance estimates $\{\hat{g}_n^{(-k)} : k \in [K]\}$ and $\{\hat{\alpha}_n^{(-k)} : k \in [K]\}$. Then, we have*

$$\sqrt{n}(\hat{V}_n - V^*) = \frac{1}{\sqrt{n}} \sum_{m=1}^n \{ \Psi^*(Z_m; g^*, \alpha^*) - V^* \} + o_{\mathbb{P}}(1).$$

where $\hat{V}_n$ is the cross-fitting estimate

$$\hat{V}_n := \frac{1}{n} \sum_{k=1}^K \sum_{i \in \mathcal{I}_k} \Psi^{\beta_n} \left(Z_i; \hat{g}_n^{(-k)}, \hat{\alpha}_n^{(-k)} \right).$$

Proof. For $k \in [K]$, let $(Z_m^{(k)})_{m \geq 1}$ denote the sequence of random variables belonging to fold $\mathcal{I}_k$. Observing that, by constructing of the folds, $\hat{g}_n^{(-k)}$ and $\hat{\alpha}_n^{(-k)}$ are independent of $Z_1^{(k)}, \dots, Z_{n(k)}^{(k)}$, where we have defined $n(k) := \lfloor n/k \rfloor$. We have from the conclusion of Theorem 4.2 that

$$\begin{aligned} \sum_{i \in \mathcal{I}_k} \Psi^{\beta_n} \left(Z_i; \hat{g}_n^{(-k)}, \hat{\alpha}_n^{(-k)} \right) &= \sum_{m=1}^{n(k)} \Psi^{\beta_n} \left(Z_m^{(k)}; \hat{g}_n^{(-k)}, \hat{\alpha}_n^{(-k)} \right) \\ &= \sum_{m=1}^{n(k)} \Psi^* \left(Z_m^{(k)}; g^*, \alpha^* \right) + o_{\mathbb{P}}(n^{1/2}) \\ &= \sum_{i \in \mathcal{I}_k} \Psi^* \left(Z_i; g^*, \alpha^* \right) + o_{\mathbb{P}}(n^{1/2}). \end{aligned}$$

From this it follows that we have

$$\sqrt{n}(\widehat{V}_n - V^*) = \frac{1}{\sqrt{n}} \sum_{k=1}^K \sum_{i \in \mathcal{I}_k} \{\Psi^*(Z_i; g^*, \alpha^*) - V^*\} + o_{\mathbb{P}}(1),$$

proving the desired result. Similarly, for showing consistency of the plug-in variance estimate, one has

$$\widehat{\Sigma}_n - \Sigma^* = \frac{1}{n} \sum_{k=1}^K \frac{n}{K} (\widehat{\Sigma}_n^{(k)} - \Sigma^*) = \frac{1}{n} \sum_{k=1}^K o_{\mathbb{P}}(n) = o_{\mathbb{P}}(1),$$

where we have defined $\widehat{\Sigma}_n^{(k)} := \frac{K}{n} \sum_{i \in \mathcal{I}_k} \left\{ \Psi^{\beta_n}(Z_i; \widehat{g}_n^{(-k)}, \widehat{\alpha}_n^{(-k)}) - \widehat{V}_n \right\}^2$ and note that $\widehat{\Sigma}_n^{(k)} - \Sigma^* = o_{\mathbb{P}}(1)$ follows from an analogous argument to the one used in the proof of Corollary 4.3, since $\widehat{V}_n \xrightarrow[n \rightarrow \infty]{\mathbb{P}} V^*$. It thus follows that

$$\sqrt{n} \widehat{\Sigma}_n^{-1/2} (\widehat{V}_n - V^*) = \frac{1}{\sqrt{n}} \sum_m (\Sigma^*)^{-1/2} \{\Psi^*(Z; g^*, \alpha^*) - V^*\} + o_{\mathbb{P}}(1),$$

thus proving the desired result. $\square$

We conclude this appendix by discussing sufficient conditions under which the exchanging of expectations and Gateaux derivatives in Condition 6 of Theorem 4.2 is justified.

Lemma C.3. *Let n be fixed and $\widehat{g}_n$ and g^* be as in Theorem 4.2 and assume the score vector $\psi(x; g) := (\psi_1(x; g_1), \dots, \psi_N(x; g_N)) \in \mathbb{R}^N$ satisfies Assumption 2. Let $\bar{g} \in [g^*, \widehat{g}_n]$ be an arbitrary and $\widehat{g}_n$ -measurable. Then, we can exchange Gateaux derivatives and expectations, i.e. we have*

$$D_g \mathbb{E}_X [\mathbf{sm}^{\beta_n} \psi(X; \bar{g})(\Delta_n)] = \mathbb{E}_X [D_g \mathbf{sm}^{\beta_n} \psi(X; \bar{g})(\Delta_n)] \quad \mathcal{E} \quad D_g^2 \mathbb{E}_X [\mathbf{sm}^{\beta_n} \psi(X; \bar{g})](\Delta_n) = \mathbb{E}_X [D_g^2 \mathbf{sm}^{\beta_n} \psi(X; \bar{g})(\Delta_n)]$$

for each $n \geq 1$, where $\Delta_n := \widehat{g}_n - g^*$.

To prove the result, we use the dominated convergence theorem for conditional expectations.

Lemma C.4. *Let $(X_m)_{m \geq 1}$ and X be random variables defined on some probability space $(\Omega, \mathcal{F}, \mathbb{P})$ such that $X_m \xrightarrow[m \rightarrow \infty]{a.s.} X$. Further suppose that $\sup_{m \geq 1} |X_m| \leq B$ almost surely for some random variable B satisfying $\mathbb{E}B < \infty$. Let $\mathcal{G} \subset \mathcal{F}$ be a sub σ -algebra. Then, we have*

$$\lim_{m \rightarrow \infty} \mathbb{E}(X_m | \mathcal{G}) = \mathbb{E}(X | \mathcal{G}) \quad \text{almost surely.}$$

Proof of Lemma C.3. We start by checking interchangeability for the first Gateaux derivative. Let $\bar{g} \in [g^*, \widehat{g}_n]$ be arbitrary and $\widehat{g}_n$ -measurable, and set $\Delta_n := \widehat{g}_n - g^*$. We have

$$\begin{aligned} D_g \mathbb{E}_X [\mathbf{sm}^{\beta_n} \psi(X; \bar{g})](\Delta_n) &= \partial_t \mathbb{E}_X [\mathbf{sm}^{\beta_n} \psi(X; \bar{g} + t\Delta_n)] \Big|_{t=0} && \text{(Definition of Gateaux Derivative)} \\ &= \lim_{t \rightarrow 0} \mathbb{E}_X \left[\frac{\mathbf{sm}^{\beta_n} \psi(X; \bar{g} + t\Delta_n) - \mathbf{sm}^{\beta_n} \psi(X; \bar{g})}{t} \right] \\ &= \lim_{t \rightarrow 0} \mathbb{E}_X \left[\frac{D_g \mathbf{sm}^{\beta_n} \psi(X; \tilde{g}_t)(t\Delta_n)}{t} \right] && \text{(Mean value theorem)} \\ &= \lim_{t \rightarrow 0} \mathbb{E}_X [D_g \mathbf{sm}^{\beta_n} \psi(X; \tilde{g}_t)(\Delta_n)] && \text{(Linearity)} \end{aligned}$$

where $\tilde{g}_t \in [\bar{g}, \bar{g} + t\Delta_n]$ exists by the mean value theorem. We aim now to show that the final limit is $\mathbb{E}_X [D_g \mathbf{sm}^{\beta_n} \psi(X; \bar{g})(\Delta_n)]$. First, note that $\bar{g} + t\Delta_n \xrightarrow[t \rightarrow 0]{a.s.} \bar{g}$ since Δ_n is almost surely bounded. Further, since $\tilde{g}_t \in [\bar{g}, \bar{g} + t\Delta_n]$, we have that $\tilde{g}_t$ also converges almost surely to $\bar{g}$. Next, note that $\tilde{g}_t$ can be written as

$$\tilde{g}_t = g^* + T_t(x)\Delta_n$$

where $T_t(x) \in [0, 2]$ for each x .⁴ Since

$$D_g \mathbf{sm}^{\beta_n} \psi(X; \tilde{g}_t)(\Delta_n) = \sum_{i=1}^N \partial_i \mathbf{sm}^{\beta_n} \psi(X; \tilde{g}_t) \cdot D_{g_i} \psi_i(X; \tilde{g}_{t,i})(\Delta_n) = \nabla_u \mathbf{sm}^{\beta_n} \psi(X; \tilde{g}_t)^\top D_g \psi(X; \tilde{g}_t)(\Delta_n), \quad (6)$$

and $\ell_{x, \Delta_n}(\epsilon) = \psi(x; g^* + \epsilon \Delta_n)$ and $\partial_\epsilon \ell_{x, \Delta_n}(\epsilon) = \partial_s \psi(x; g^* + s \Delta_n) \Big|_{s=\epsilon} = D_g \psi(x; g + \epsilon \Delta_n)(\Delta_n)$ are continuous in ϵ for each fixed x and Δ_n by assumption, it follows that $D_g \mathbf{sm}^{\beta_n} \psi(X; \tilde{g}_t)(\Delta_n) \rightarrow D_g \mathbf{sm}^{\beta_n} \psi(X; \bar{g})(\Delta_n)$ almost surely as $t \rightarrow 0$. Next, noting that $|T_t(x)| \leq 2$ for each x and t , we can apply Condition 6/Assumption 2 to guarantee the existence of a random $F_{\Delta_n, C}$ (with $C = 2$) satisfying (1) $\mathbb{E}[F_{\Delta_n, C}] < \infty$ and (2) $\left\| \partial_\epsilon \psi(X; g^* + \epsilon \Delta_n) \Big|_{\epsilon=T_t(x)} \right\|_2 \leq F_{\Delta_n, C}$ almost surely. Putting things together, we have

$$\begin{aligned} |D_g \mathbf{sm}^{\beta_n} \psi(X; \tilde{g}_t)(\Delta_n)| &= \sum_{i=1}^N \partial_i \mathbf{sm}^{\beta_n} \psi(X; \tilde{g}_t) \cdot D_{g_i} \psi_i(X; \tilde{g}_{t,i})(\Delta_n) && \text{(Equation (6))} \\ &\leq \|\nabla_u \mathbf{sm}^{\beta_n} \psi(X; \tilde{g}_t)\|_2 \|D_g \psi(X; \tilde{g}_t)(\Delta_n)\|_2 && (\ell_\infty / \ell_1 \text{ Cauchy-Schwarz}) \\ &\leq \sqrt{N} \|\nabla_u \mathbf{sm}^{\beta_n} \psi(X; \tilde{g}_t)\|_\infty \left\| \partial_\epsilon \psi(X; g^* + \epsilon \Delta_n) \Big|_{\epsilon=T_t(x)} \right\|_2 \\ &\leq C_1 F_{\nu, C}, \end{aligned}$$

which follows for some absolute constant $C_1 > 0$ from by the boundedness of the ℓ_∞ norm of the softmax function (via Lemma A.1). Thus, noting that $\mathbb{E}_X[D_g \mathbf{sm}^{\beta_n} \psi(X; \tilde{g}_t)(\Delta_n)] = \mathbb{E}[D_g \mathbf{sm}^{\beta_n} \psi(X; \tilde{g}_t)(\Delta_n) \mid \hat{g}_n]$, an application of the dominated convergence theorem for conditional expectations (Lemma C.4) yields the desired result.

Next, we show interchangeability for the second derivative. First, note that we can write

$$\begin{aligned} D_g^2 \mathbb{E}_X [\mathbf{sm}^{\beta_n} \psi(X; \bar{g})] (\Delta_n) &= \partial_t^2 \mathbb{E}_X [\mathbf{sm}^{\beta_n} \psi(X; \bar{g} + t \Delta_n)] \Big|_{t=0} \\ &= \lim_{t \rightarrow 0} \mathbb{E}_X \left[\frac{\mathbf{sm}^{\beta_n} \psi(X; \bar{g} + t \Delta_n) - 2\mathbf{sm}^{\beta_n} \psi(X; \bar{g}) + \mathbf{sm}^{\beta_n} \psi(X; \bar{g} - t \Delta_n)}{t^2} \right] \\ &= \lim_{t \rightarrow 0} \mathbb{E}_X \left[\frac{D_g^2 \mathbf{sm}^{\beta_n} \psi(X; \tilde{g}_t)(t \Delta_n, t \Delta_n)}{t^2} \right] \\ &= \lim_{t \rightarrow 0} \mathbb{E}_X [D_g^2 \mathbf{sm}^{\beta_n} \psi(X; \tilde{g}_t)(\Delta_n)], \end{aligned}$$

where the first equality comes from an equivalent characterization of the second derivative and the second equality comes from the mean value theorem for some $\tilde{g}_t \in [\bar{g} - t \Delta_n, \bar{g} + t \Delta_n]$. Again, we can write $\tilde{g}_t = g^* + T_t(x) \Delta_n$ for some $T_t(x) \in [-2, 2]$. At this point, we can use the fact that

$$\begin{aligned} D_g^2 \mathbf{sm}^{\beta_n} \psi_\ell(X; \tilde{g}_t)(\Delta_n) &= \underbrace{D_g \psi(X; \tilde{g}_t)(\Delta_n)^\top \nabla_u^2 \mathbf{sm}^{\beta_n} \psi(X; \tilde{g}_t) D_g \psi(X; \tilde{g}_t)(\Delta_n)}_{T_1} \\ &\quad + \underbrace{\nabla_u \mathbf{sm}^{\beta_n} \psi(X; \tilde{g}_t)^\top D_g^2 \psi(X; \tilde{g}_t)(\Delta_n)}_{T_2} \end{aligned}$$

from the proof of Lemma A.1. From the above decomposition, the fact that $\tilde{g}_t = g^* + T_t(x) \Delta_n \rightarrow \bar{g}$ almost surely, and the continuity of $\psi(x; g^* + \epsilon \nu)$, $\partial_s \psi(x; g^* + s \nu) \Big|_{s=\epsilon}$, and $\partial_\epsilon^2 \psi(x; g^* + s \nu) \Big|_{s=\epsilon}$ for any x, ν , we have that $D_g^2 \mathbf{sm}^{\beta_n} \psi(X; \tilde{g}_t) \xrightarrow[t \rightarrow 0]{a.s.} D_g^2 \mathbf{sm}^{\beta_n} \psi(X; \bar{g})$. Now, we show $|T_1|$ and $|T_2|$ are bounded above by integrable random variables that do not depend on t . First, we have

$$|T_1| \leq \|\nabla_u^2 \mathbf{sm}^{\beta_n} \psi(X; \tilde{g}_t)\|_{op} \|D_g \psi(X; \tilde{g}_t)(\Delta_n)\|_2 = \|\nabla_u^2 \mathbf{sm}^{\beta_n} \psi(X; g)\|_{op} \left\| \partial_\epsilon \psi(X; g^* + \epsilon \Delta_n) \Big|_{\epsilon=T_t(x)} \right\|_2 \leq C_{2,n} F_{\Delta_n, C},$$

⁴This is because we first have that $\bar{g} = g^* + T' \Delta_n$, where T' is some $\hat{g}_n$ -measurable random variable in $[0, 1]$, and then $\tilde{g} = \bar{g} + T'(x) \Delta_n = g^* + (T' + T'(x)) \Delta_n$, where $T'(x) \in [0, 1]$ for each x .

which follows from the fact that, for fixed n , the operator norm of the soft-max is $O(\beta_n)$ (Lemma A.1 Part 3) and from the fact that we assume the first Gateaux derivative is dominated. Note that while $C_{2,n}$ does depend on n (via β_n) it importantly does not depend on t . A similar analysis yields

$$|T_2| \leq \|\nabla_u \mathbf{sm}^{\beta_n} \psi(X; \tilde{g}_t)\|_2 \|D_g^2 \psi(X; \tilde{g}_t)(\Delta_n)\|_2 \leq \sqrt{N} \|\nabla_u \mathbf{sm}^{\beta_n} \psi(X; \tilde{g}_t)\|_\infty \|\partial_\epsilon^2 \psi(X; g^* + \epsilon \Delta_n)|_{\epsilon=T_t(x)}\|_2 \leq C_1 F_{\Delta_n, C}.$$

where C_1 is as above. Thus, since both T_1 and T_2 are absolutely bounded, $D_g^2 \mathbf{sm}^{\beta_n} \psi(X; \tilde{g}_t)(\Delta_n)$ is as well. The interchangeability of derivation and expectation now once again follows from Lemma C.4. $\square$

C.1 Checking Conditions for Balke and Pearl Bounds

We now check that the scores outlined in Subsection 4.1 satisfy the conditions of Theorem 4.2.

Conditions 1, 2, and 4 of the theorem are simply assumptions that one must place on the density of the sub-optimality gaps, the smoothing parameters, and the nuisance learning rates. Condition 3 is straightforward to verify since we have assumed strong positivity (i.e. $p^*(x) \in [\eta, 1 - \eta]$). In particular, we have

$$\begin{aligned} \|\psi_k(X; q_k) - \psi_k(X; q'_k)\|_{L^2(P_X)}^2 &= \mathbb{E} \left[\left(\sum_{(y,a,v)} \text{sgn}(y, a, v) \{q_{ya}(x, v) - q'_{ya}(x, v)\} \right)^2 \right] && \text{(Definition of } \psi_k) \\ &\lesssim \sum_{(y,a,v) | \text{sgn}(y,a,v)=1} \mathbb{E} \left[(q_{ya}(X, v) - q'_{ya}(X, v))^2 \right] && \text{(Parallelogram Law)} \\ &\lesssim \sum_{\substack{(y,a,v) \\ |\text{sgn}(y,a,v)=1}} \mathbb{E} \left[\frac{\mathbb{1}\{V=v\}}{p^*(X)} \{q_{ya}(X, V) - q'_{ya}(X, V)\}^2 \right] && \text{(Conditional Ignorability of } V) \\ &\lesssim \sum_{\substack{(y,a,v) \\ |\text{sgn}(y,a,v)=1}} \mathbb{E} \left[\{q_{ya}(X, V) - q'_{ya}(X, V)\}^2 \right] && \text{(Strong Positivity)} \\ &= \|q_k - q'_k\|_{L^2(P_W)}^2 \end{aligned}$$

Analogously, Condition 5 will hold as U_k and q_ℓ^* are bounded by definition, positivity ensures ζ_ℓ^* and thus $\alpha_\ell^{\beta_n}$ are bounded, and the boundedness of $\hat{g}_n$ and $\hat{\alpha}_n$ depend on the choice of learning algorithms. Lastly, one can directly compute for all $\bar{q}$ and any $\hat{q}_n$ that

$$\mathbb{E}[D_{q_k} \psi_k(X; \bar{q}_k)(\hat{q}_{n,k} - q_k^*)^2] = \|\psi_k(X; \hat{q}_{n,k}) - \psi_k(X; q_k^*)\|_{L^2(P_X)}^2 \lesssim \|\hat{q}_{n,k} - q_k^*\|_{L^2(P_X)}^2.$$

and that $\mathbb{E}[D_q^2 \psi(X; \bar{q})(\hat{q}_n - q^*)] = 0$, since each ψ_k is affine in q_k .⁵ Lastly, since $\hat{q}_{n,k}$ and q_k^* are almost surely bounded, the random variable $F_{\hat{q}_n - q^*, C}$ is also bounded and hence integrable. Thus, the final condition (Condition 6) is also satisfied.

D A Smoothed Central Limit Theorem

In this appendix, we provide a simple but useful central limit theorem (CLT) for target parameters that are specified as the expectation of a sequence of smoothed scores. We note that other such CLTs have been proven in the causal inference literature. For instance, Chen et al. [2023] (in Theorem 8 of their paper) prove such a result for smoothed scores that are Neyman orthogonal with respect to infinite-dimensional nuisance parameters but not orthogonal with respect to finite-dimensional ones. Unfortunately, we cannot directly apply this result to our setting, as the authors assume (in Assumption 4 of their work) that the second order Gateaux derivatives of the smoothed score with respect to the infinite-dimensional nuisances are uniformly bounded over choice of smoothing parameters β . In our setting, this Hessian will scale linearly with β , thus preventing direct application.

⁵Here, we note that each q_k is a vector-valued function of dimension $d_k = |\mathcal{I}_k|$. We can thus index coordinates of q_k by tuples $(q, a) \in \mathcal{I}_k$.

Theorem D.1. Suppose, for any $\beta > 0$, there is a score $m^\beta(Z; g)$. Define $\theta_\beta := \mathbb{E}m^\beta(Z; g^{\beta_n})$, where g^β is some true nuisance parameter. Let $(\beta_n)_{n \geq 0}$ satisfy $\beta_n \rightarrow \infty$, let $(Z_n)_{n \geq 0}$ be a sequence of i.i.d. draws from some distribution $\mathbb{P}_Z$, let $(\hat{g}_n)$ be a sequence of random nuisance estimates independent of (Z_n) , and define

$$\hat{\theta}_n := \mathbb{P}_n m^{\beta_n}(Z; \hat{g}_n).$$

Suppose the following hold:

1. (Neyman Orthogonality) For any $\beta > 0$, the score m^β is Neyman orthogonal, i.e. for all $g \in L^2(P_W)$

$$D_g \mathbb{E}_Z [m^\beta(Z; g^\beta)](g - g^\beta) = 0.$$

2. (Second Order Gateaux Derivatives) The second-order errors shrink sufficiently quickly, i.e. $\forall \bar{g} \in [g^{\beta_n}, \hat{g}]$

$$D_g^2 \mathbb{E}_Z [m^\beta(Z; \bar{g})] (\hat{g}_n - g^{\beta_n}) = o_{\mathbb{P}}(n^{-1/2}).$$

3. (Convergence of Score) We have, as $\beta \rightarrow \infty$,

$$m^\beta(Z; g^\beta) \xrightarrow{\mathbb{P}} m^*(Z; g^*)$$

for some limiting score m_* and some nuisance g_0 .

4. (Stochastic Equicontinuity) We have

$$|\mathbb{G}_n m^{\beta_n}(Z; \hat{g}_n) - \mathbb{G}_n m^{\beta_n}(Z; g^{\beta_n})| = o_{\mathbb{P}}(1),$$

where $\mathbb{G}_n := \sqrt{n}(\mathbb{P}_n - \mathbb{E}_Z)$.

5. (Regularity) There exists $D > 0$ and $\epsilon > 0$ such that, for any $\beta > 0$, we have

$$\|m^\beta(Z; g^\beta)\|_{L^{2+\epsilon}(P_Z)} \leq D.$$

Further, we assume $\|m^*(Z; g^*)\|_{L^{2+\epsilon}(P_Z)} \leq D$ as well.

Then, we have the following asymptotic linearity result:

$$\sqrt{n}(\hat{\theta}_n - \theta^{\beta_n}) = \frac{1}{\sqrt{n}} \sum_{m=1}^n (m^*(Z; g^*) - \theta^*) + o_{\mathbb{P}}(1).$$

In particular, this implies

$$\sqrt{n}(\hat{\theta}_n - \theta^{\beta_n}) \Rightarrow \mathcal{N}(0, \Sigma^*)$$

where $\Sigma^* := \text{Var}[m^*(Z; g^*)]$.

Before proving the result, we state Vitali's theorem, a result that relates convergence of a sequence of random variables in L^p to convergence in probability and uniform integrability.

Proposition D.2 (Bogachev and Ruas [2007]). Suppose $(X_n)_{n \geq 1}$ are a sequence of random variables and X is a random variable in a measure space $(\Omega, \mathcal{F}, P)$ such $X_n \in L^p(P)$ for each n and $X \in L^p(P)$. The following are equivalent.

1. $\|X_n - X\|_{L^p(P)} \xrightarrow{n \rightarrow \infty} 0$, i.e. $(X_n)_{n \geq 0}$ converge in L^p to X .

2. $X_n \xrightarrow[n \rightarrow \infty]{\mathbb{P}} X$ and $(|X_n|^p)_{n \geq 1}$ is uniformly integrable.⁶

⁶We say a collection of random variables $(X_\alpha)_{\alpha \in A}$ is uniformly integrable if, for any $\delta > 0$, there exists a constant $K_\delta > 0$ such that

$$\sup_{\alpha \in A} \mathbb{E}[|X_\alpha| \mathbb{1}\{|X_\alpha| > K_\delta\}] \leq \delta.$$

We also use the following result which can be used to relate uniformly bounded higher moments of a sequence of random variables/scores to uniform integrability.

Proposition D.3 (Poussin [1915], Dellacherie and Meyer [2011]). *A sequence of integrable random variables $(X_n)_{n \geq 1}$ is uniformly integrable if and only if there is an increasing convex function $\Phi : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ such that*

$$\lim_{t \rightarrow \infty} \frac{\Phi(t)}{t} = \infty \quad \text{and} \quad \sup_n \mathbb{E}[\Phi(|X_n|)] < \infty.$$

Proof of Theorem D.1. First, we note that we have the following decomposition.

$$\begin{aligned} \sqrt{n}(\hat{\theta}_n - \theta^{\beta_n}) &= \sqrt{n} \{ \mathbb{P}_n m^{\beta_n}(Z; \hat{g}_n) - \mathbb{E}_Z m^{\beta_n}(Z; g^{\beta_n}) \} \\ &= \sqrt{n} \{ \mathbb{P}_n m^{\beta_n}(Z; \hat{g}_n) \pm \mathbb{P}_n m^{\beta_n}(Z; g^{\beta_n}) \pm \mathbb{E}_Z m^{\beta_n}(Z; \hat{g}_n) \pm \mathbb{E}_Z m^{\beta_n}(Z; g^{\beta_n}) - \mathbb{E}_Z m^{\beta_n}(Z; g^{\beta_n}) \} \\ &= \sqrt{n} \{ \mathbb{P}_n m^{\beta_n}(Z; \hat{g}_n) - \mathbb{E}_Z m^{\beta_n}(Z; \hat{g}_n) - \mathbb{P}_n m^{\beta_n}(Z; g^{\beta_n}) + \mathbb{E}_Z m^{\beta_n}(Z; g^{\beta_n}) \} \\ &\quad + \sqrt{n} \{ \mathbb{P}_n m^{\beta_n}(Z; g^{\beta_n}) - \mathbb{E}_Z m^{\beta_n}(Z; g^{\beta_n}) \} + \sqrt{n} \{ \mathbb{E}_Z m^{\beta_n}(Z; \hat{g}_n) - \mathbb{E}_Z m^{\beta_n}(Z; g^{\beta_n}) \} \\ &= \underbrace{\mathbb{G}_n m^{\beta_n}(Z; \hat{g}_n) - \mathbb{G}_n m^{\beta_n}(Z; g^{\beta_n})}_{T_1} + \underbrace{\mathbb{G}_n m^{\beta_n}(Z; g^{\beta_n})}_{T_2} + \underbrace{\mathbb{E}_Z m^{\beta_n}(Z; \hat{g}_n) - \mathbb{E}_Z m^{\beta_n}(Z; g^{\beta_n})}_{T_3}. \end{aligned}$$

We handle each of the three terms separately. Handling the first term follows by assumption, as we have by Condition 4

$$T_1 = \mathbb{G}_n m^{\beta_n}(Z; \hat{g}_n) - \mathbb{G}_n m^{\beta_n}(Z; g^{\beta_n}) = o_{\mathbb{P}}(1).$$

Handling follows from the assumptions regarding Neyman orthogonality and bounded second order errors:

$$\begin{aligned} \frac{1}{\sqrt{n}} T_3 &= \mathbb{E}_Z m^{\beta_n}(Z; \hat{g}_n) - \mathbb{E}_Z m^{\beta_n}(Z; g^{\beta_n}) \\ &= D_g \mathbb{E}_Z [m^{\beta_n}(Z; g^{\beta_n})](\hat{g}_n - g^{\beta_n}) + \frac{1}{2} D_g^2 \mathbb{E}_Z [m^{\beta_n}(Z; \bar{g})](\hat{g}_n - g^{\beta_n}) \\ &= \frac{1}{2} D_g^2 \mathbb{E}_Z [m^{\beta_n}(Z; \bar{g})](\hat{g}_n - g^{\beta_n}) \quad (\text{Condition 1}) \\ &= o_{\mathbb{P}}(n^{-1/2}) \quad (\text{Condition 2}), \end{aligned}$$

so multiplying everything by $\sqrt{n}$ yields that $T_3 = o_{\mathbb{P}}(1)$.

The last thing to do is analyze T_2 .

We can rewrite T_2 as follows:

$$T_2 = \mathbb{G}_n m^*(Z; g^*) + \{ \mathbb{G}_n m^{\beta_n}(Z; g^{\beta_n}) - \mathbb{G}_n m^*(Z; g^*) \}.$$

We need to show the last term is $o_{\mathbb{P}}(1)$. We can control this term by noting

$$\begin{aligned} \mathbb{E} | \mathbb{G}_n m^{\beta_n}(Z; g^{\beta_n}) - \mathbb{G}_n m^*(Z; g^*) |^2 &= \text{Var}[m^{\beta_n}(Z; g^{\beta_n}) - m^*(Z; g^*)] \\ &\leq \mathbb{E} | m^{\beta_n}(Z; g^{\beta_n}) - m^*(Z; g^*) |^2 \\ &= o(1). \end{aligned}$$

We now justify the final equality. In particular, we leverage the two propositions stated before the proof to show that

$$\lim_{n \rightarrow \infty} \mathbb{E} | m^{\beta_n}(Z; g^{\beta_n}) - m^*(Z; g^*) |^2 = 0.$$

We want to this by applying Vitali's Theorem (Proposition D.2). Since we already know that $m^{\beta_n}(Z; g^{\beta_n}) - m^*(Z; g^*) = o_{\mathbb{P}}(1)$ by Condition 3, it suffices to check that the sequence $(m^{\beta_n}(Z; g^{\beta_n})^2)_{n \geq 1}$ is uniformly integrable. To see this, we take $\Phi(t) := t^{1+\frac{\epsilon}{2}}$. Then, we have the identity

$$\sup_{n \geq 1} \mathbb{E} [\Phi(|m^{\beta_n}(Z; g^{\beta_n})|)] = \sup_{n \geq 1} \mathbb{E} [|m^{\beta_n}(Z; g^{\beta_n})|^{2+\epsilon}] \leq D,$$

where the final inequality follows from Condition 3. Thus the sequence is uniformly integrable by Proposition D.3. From this, we see that we have

$$\mathbb{P}(|\mathbb{G}_n m^{\beta_n}(Z; g^{\beta_n}) - \mathbb{G}_n m^*(Z; g^*)| \geq \delta) \leq \frac{\mathbb{E} |\mathbb{G}_n m^{\beta_n}(Z; g^{\beta_n}) - \mathbb{G}_n m^*(Z; g^*)|^2}{\delta} = o(1)$$

for any $\delta > 0$. This completes the proof. $\square$

E A Smoothing Estimator Due to Park

In this appendix, we discuss a smoothing-based estimator for the average welfare under the optimal treatment policy due to Park [2024]. Interestingly, while the authors use a smoothed, de-biased score that is similar to the one considered in this paper, they obtain asymptotic normality at slower than $\sqrt{n}$ convergence rates. In particular, they obtain $\sqrt{n/\beta_n^2}$ convergence rates, where (β_n) is a sequence of chosen smoothing parameters tending towards infinity. We argue that, when one scales the difference between the statistic and true parameter by the additional factor of β_n^{-1} in many settings, the asymptotic variance is actually zero, preventing the construction of confidence intervals. We now dive into more detail on the setup, statistics, and result.

The authors consider the setting of binary treatments, and in particular attempt to estimate the difference in expected welfare between the optimal treatment policy and the constant treatment policy where all units are kept under control, i.e.

$$V^* := \mathbb{E}[\tau^*(X) \mathbb{1}\{\tau^*(X) \geq 0\}],$$

which is similar (modulo an addition of $Q^*(0, X)$) to the value of the optimal treatment policy as formulated in Remark B.1 in Appendix B. As in our work, they propose approximating the above, non-differentiable score by the following softmax approximation

$$V^\beta := \mathbb{E}[\tau^*(X) \sigma(\beta \tau^*(X))] = \mathbb{E} \varphi_\beta(\tau^*(X)),$$

where we define $\varphi_\beta(u) := u \sigma(\beta u)$ for notational ease. Further, they correspondingly define the de-biased smoothed score as

$$\Psi^\beta(z; Q, \alpha) := \tau(x) \sigma(\beta \tau(x)) + \sum_{a \in \{0,1\}} \alpha_a(a, x) (y - Q(a, x)),$$

where $Q^*(a, x) := \mathbb{E}[Y \mid X = x, A = a]$ and $\tau^*(x) = Q^*(1, x) - Q^*(0, x)$. One can compute that the true Riesz representers α_1^β and α_0^β are given as

$$\begin{aligned} \alpha_0^\beta(a, x) &= -\varphi'_\beta(\tau^*(x)) \frac{1-a}{1-p^*(x)} \\ \alpha_1^\beta(a, x) &= \varphi'_\beta(\tau^*(x)) \frac{a}{p^*(x)}, \end{aligned}$$

although we note that the authors do not provide an explicit characterization of the representers in their works.

The authors in Park [2024] claim (in Proposition 1) that, under standard assumptions (Assumptions 1 - 5 in the cited work), letting $\widehat{V}_n := \sum_{k=1}^K \sum_{i \in \mathcal{I}_k} \Psi^{\beta_n}(Z; \widehat{Q}_n^{(-k)}, \widehat{\alpha}_n^{(-k)})$ be a cross-fit estimate, one has

$$\sqrt{\frac{n}{\beta_n^2}} (\widehat{V}_n - V^{\beta_n}) \Rightarrow \mathcal{N}(0, \Sigma) \quad \text{where} \quad \Sigma := \frac{1}{16} \text{Var}[\tau^*(X)^2].$$

Interestingly, this asymptotic variance does not depend on the first-order debiasing correction terms. In their derivation of the above limiting result, the authors attempt to prove the following identity

$$\lim_{n \rightarrow \infty} \text{Var} \left[\frac{1}{\beta_n} \Psi^{\beta_n}(Z; Q^*, \alpha^{\beta_n}) \right] = \frac{1}{16} \text{Var}[\tau^*(X)^2].$$

This is somewhat unintuitive, as based on the estimators presented in Luedtke and Van Der Laan [2016] and Chen et al. [2023], we would expect $\sqrt{n}$ rates of convergence, not the slower $\sqrt{n/\beta_n^2}$ rates obtained in this result. We show that, under the mild assumption that $\mathbb{E}Q^*(a, X)^2 < \infty$, $\mathbb{E}Y^2 < \infty$, $\eta \leq p^*(X) \leq 1 - \eta$ for each $a \in \{0, 1\}$, and that $(\beta_n)_{n \geq 1}$ is a deterministic sequence, that we in fact have

$$\lim_{n \rightarrow \infty} \text{Var} \left[\frac{1}{\beta_n} \Psi^{\beta_n}(Z; Q^*, \alpha^{\beta_n}) \right] = 0, \tag{7}$$

which shows that the asymptotic normality result in Proposition 1 of Park [2024] cannot generally hold.

Proof of Equation (7). Observe first that we have the upper bound:

$$\begin{aligned} \text{Var} [\Psi^{\beta_n}(Z; Q^*, \alpha^{\beta_n})] &\leq \mathbb{E} \left[\left(\varphi_{\beta_n}(\tau^*(X)) + \sum_{a \in \{0,1\}} \alpha_a^{\beta_n}(a, X)(Y - Q^*(a, X)) \right)^2 \right] \\ &\leq \underbrace{2 \mathbb{E} \tau^*(X)^2}_{T_1} + 4 \underbrace{\sum_{a \in \{0,1\}} \mathbb{E} [(\alpha_a^{\beta_n}(a, X))^2 (Y - Q^*(a, X))^2]}_{T_2}. \end{aligned}$$

The first inequality follows from the fact that $\text{Var}[X] \leq \mathbb{E}X^2$ for all random variables X and the second inequality comes from applying the parallelogram inequality twice. We bound T_1 first. We have by assumption that

$$T_1 = \mathbb{E} \tau^*(X)^2 = \mathbb{E}[(Q^*(1, X) - Q^*(0, X))^2] \leq 2\mathbb{E}Q^*(1, X)^2 + 2\mathbb{E}Q^*(0, X)^2 \leq C_1 < \infty$$

for some absolute constant $C_1 > 0$ that doesn't depend on β . Next, we bound T_2 . In particular, we just bound the term in the sum where $a = 1$ — the other term can be analogously bounded. We have

$$\begin{aligned} \mathbb{E} [(\alpha_1^{\beta_n}(1, X))^2 (Y - Q^*(1, X))^2] &= \mathbb{E} \left[\frac{1}{p^*(X)^2} (\varphi'_\beta(\tau^*(X)))^2 (Y - Q^*(1, X))^2 \right] \\ &\leq \frac{25}{16\eta^2} \mathbb{E} [(Y - Q^*(1, X))^2] \\ &\leq \frac{25}{8\eta^2} \mathbb{E} Y^2 + \frac{25}{8\eta^2} \mathbb{E} Q^*(1, X)^2 \\ &\leq C_{2,1}, \end{aligned}$$

where the first equality follows from definition, the second follows from the fact that $p^*(X) \geq \eta$ and $\sup_{\beta > 0} \sup_u |\varphi'_\beta(u)| \leq \frac{5}{4}$, and the third inequality follows from an application of the parallelogram inequality. One can analogously obtain a bound $C_{2,0}$ on

$$\mathbb{E} [(\alpha_0^{\beta_n}(X))^2 (Y - Q^*(0, X))^2] = \mathbb{E} \left[\frac{1}{p^*(X)^2} (\varphi'_\beta(\tau^*(X)))^2 (Y - Q^*(1, X))^2 \right]$$

that doesn't depend on β . Thus, in total, we have

$$\text{Var} [\Psi^{\beta_n}(Z; Q^*, \alpha^{\beta_n})] \leq 2C_1 + 4C_{2,0} + 4C_{2,1}.$$

This immediately yields our claim as we have

$$\lim_{n \rightarrow \infty} \text{Var} \left[\frac{1}{\beta_n} \Psi^{\beta_n}(Z; Q^*, \alpha^{\beta_n}) \right] = \lim_{n \rightarrow \infty} \frac{1}{\beta_n^2} \text{Var} [\Psi^{\beta_n}(Z; Q^*, \alpha^{\beta_n})] \leq C \lim_{n \rightarrow \infty} \frac{1}{\beta_n^2} = 0.$$

□